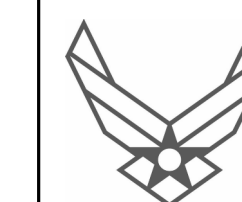


1/4/2019 10:22:50 AM



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850.438.9661
Engineering Business: EB-000034

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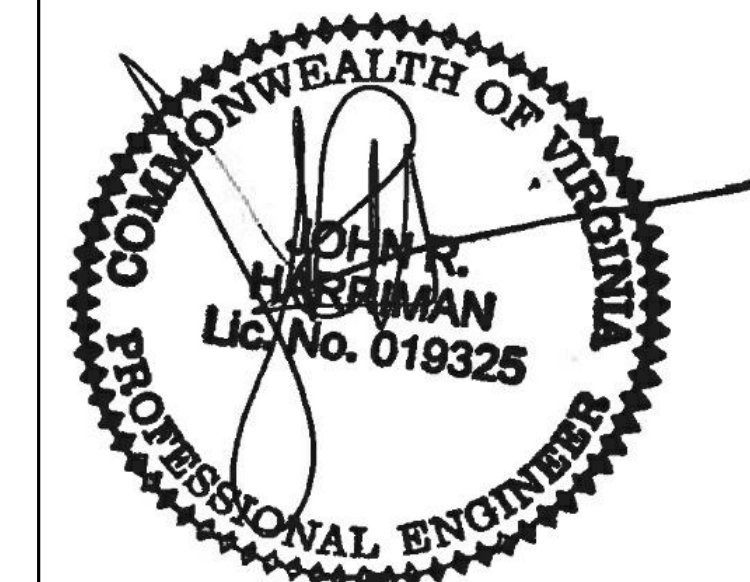
IMEG Corporation
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2882 106th Street
Suite 100
Des Moines, Iowa 50322
515.334.9906

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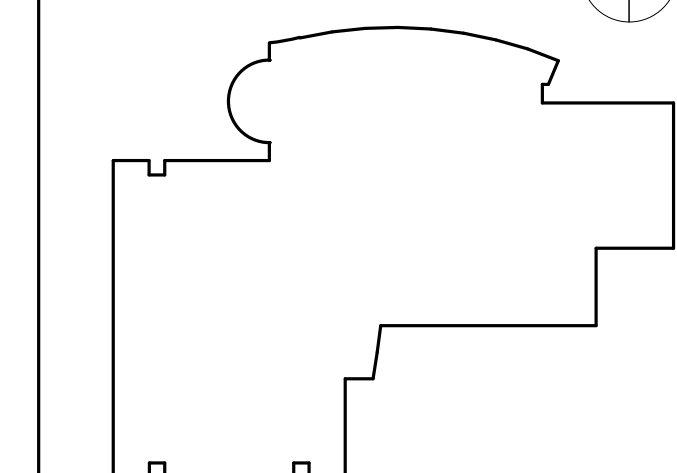
M234	PROVIDE WALL MOUNTED CO2 SENSOR. TYPICAL THROUGHOUT.
M235	PROVIDE WALL MOUNTED THERMOSTAT. TYPICAL THROUGHOUT.
M236	PROVIDE DIFFERENTIAL PRESSURE SENSORS. SEE CONTROL DRAWINGS FOR ADDITIONAL INFORMATION.

CONTRACT DOCUMENTS	01.04.
DESIGN DEVELOPMENT	11.08.
SCHEMATIC DESIGN	8.17.2

SEALS AND SIGNATURES



KEYPLAN

PLAN
NORTH

DRAWING TITLE

HVAC ZONE LEGEND

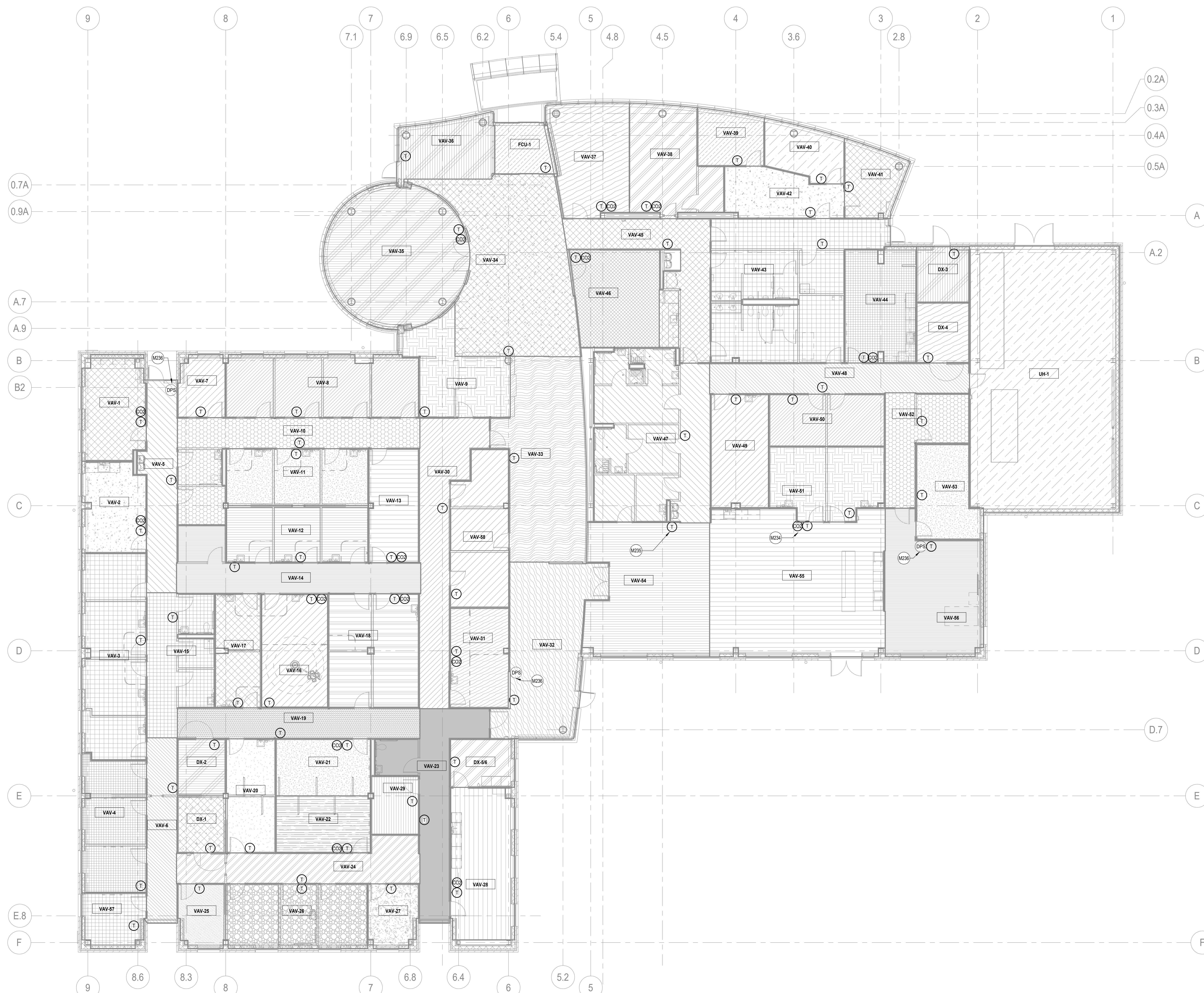
1/8" = 1'-0"

SCALE

PROJECT NUMBER

M2.1

DRAWING NUMBER



1 HVAC ZONE LEGEND
SCALE: 1/8" = 1'-0"



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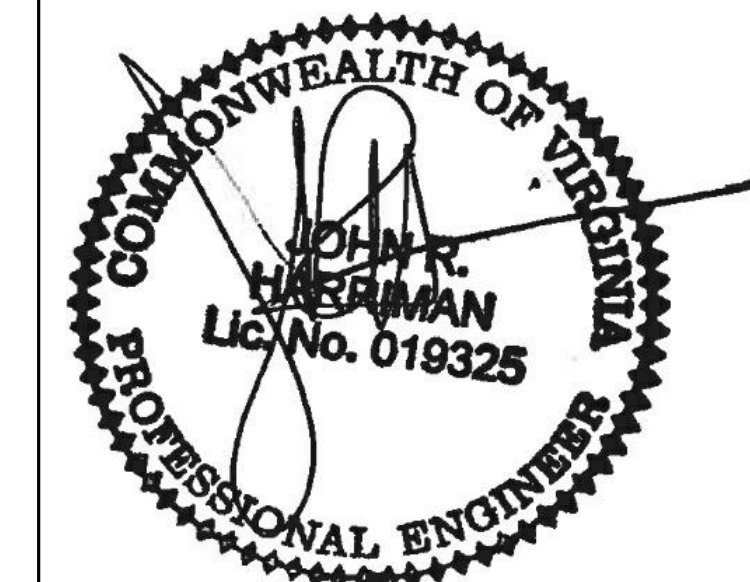
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515.254.8900

M207	18"X18" EXHAUST DUCT UP TO EF-3 ON ROOF.
M208	16"X16" EXHAUST DUCT UP TO EF-2 ON ROOF.
M209	16"X16" EXHAUST DUCT UP TO EF-1 ON ROOF.
M211	PROVIDE EMERGENCY SHUTOFF SWITCH FOR ENTIRE HVAC SYSTEM. INSTALL 48" ABOVE FINISHED FLOOR. SEE SHEET M8.8 FOR MORE INFORMATION.
M230	PROVIDE OPEN END DUCT WITH VOLUME DAMPER FOR RETURN FROM CENTRAL PARK.

ISSUED FOR	REV	DATE
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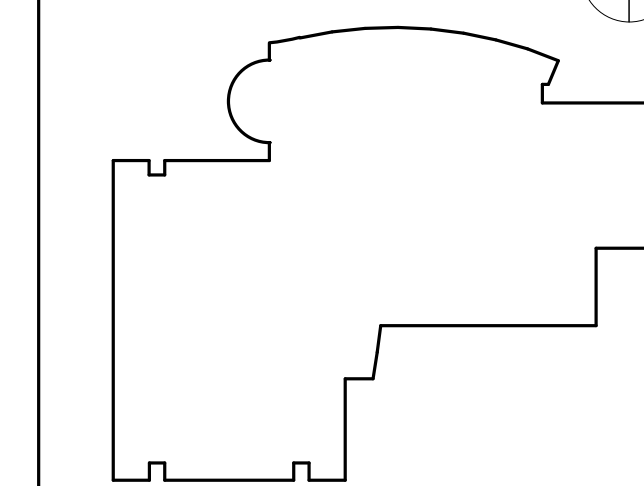
CONTRACT DOCUMENTS	01.04.
DESIGN DEVELOPMENT	11.08.
SCHEMATIC DESIGN	8.17.2

SEALS AND SIGNATURES



KEYPLAN

PLAN
NOTES



DRAWING TITLE

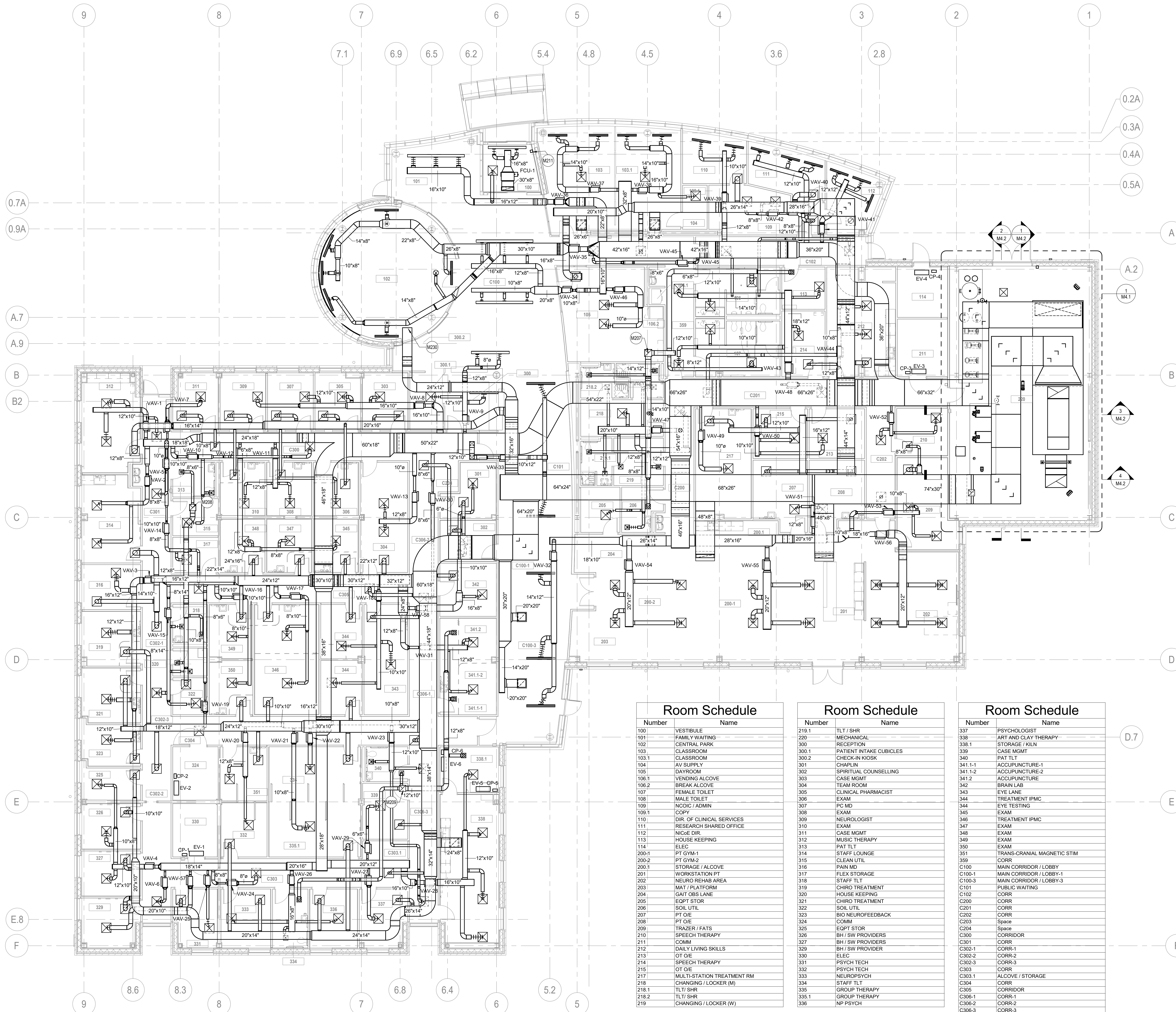
GROUND FLOOR DUCT PLAN

SCALE $1/8" = 1'-0"$

PROJECT NUMBER

M2.3

DRAWING NUMBER

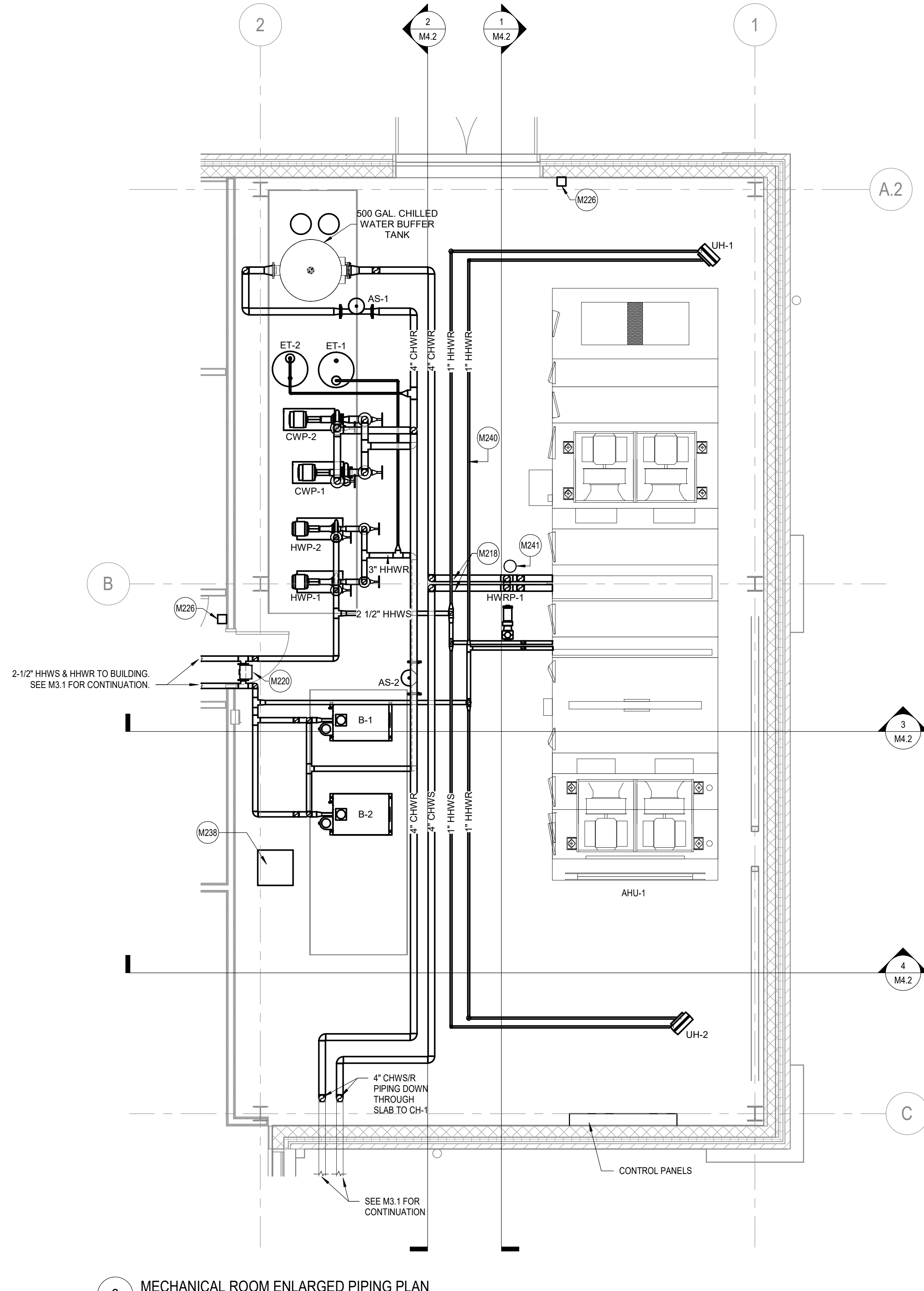
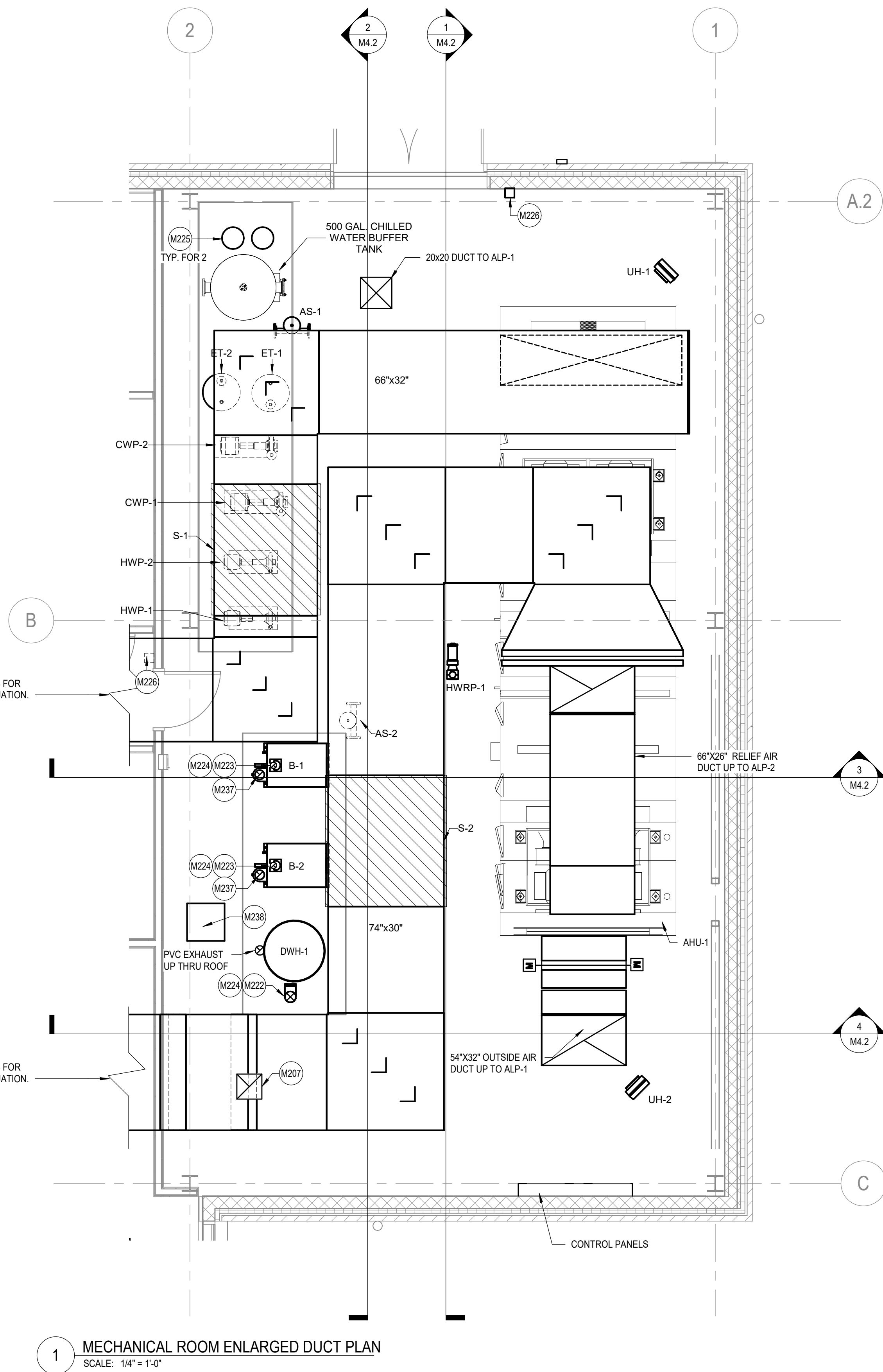


1 GROUND FLOOR DUCT PLAN
SCALE: 1/8" = 1'-0"

Plot Date: 1/4/2019 10:23:09 AM



DRAWING NUMBER



SHEET NOTES

1. PROVIDE VOLUME DAMPERS ON ALL DUCT TAKEOFFS SERVING SUPPLY, EXHAUST AND RETURN AIR TERMINALS. TYPICAL THROUGHOUT.
2. PROVIDE REMOTE DAMPER OPERATORS FOR ALL AIR TERMINALS LOCATED IN THE WOOD CEILING.
3. PROVIDE ALL RECTANGULAR MITERED ELBOWS WITH TURNING VANES.
4. PROVIDE DUCT PRESSURE READING PORTS BEFORE AND AFTER DUCT SILENCERS. TYPICAL FOR ALL DUCT SILENCERS.
5. PROVIDE DUCT DP SENSOR LOCATED IN COMPLIANCE WITH ASHRAE 90.1. STATIC PRESSURE SENSORS SHALL BE PLACED IN A POSITION SUCH THAT THE CONTROLLER SETPOINT IS NO GREATER THAN ONE-THIRD THE TOTAL STATIC PRESSURE. IF THIS RESULTS IN THE SENSOR BEING LOCATED DOWNSTREAM OF MAJOR DUCT SPLITS, MULTIPLE SENSORS SHALL BE INSTALLED IN EACH MAJOR BRANCH TO ENSURE THAT STATIC PRESSURE CAN BE MAINTAINED IN EACH.
6. IN ADDITION TO HANGERS AND SUPPORTS FOR DUCTWORK, PIPING AND EQUIPMENT, PROVIDE SEISMIC RESTRAINTS IN CONFORMANCE WITH ASCE/SEI 07-10. FOR ADDITIONAL INFORMATION, SEE SPECIFICATIONS, STRUCTURAL DRAWINGS AND CHAPTER 13 OF ASCE/SEI 07-10.



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KEYED NOTES

- | | |
|------|--|
| M207 | 18"X18" EXHAUST DUCT UP TO EF-3 ON ROOF. |
| M218 | PROVIDE DP SENSOR. SEE CONTROLS DRAWINGS. |
| M220 | PROVIDE TWO-WAY BYPASS CONTROL VALVE. |
| M222 | PROVIDE PVC INTAKE FOR DOMESTIC WATER HEATER. RUN DUCT UP THRU ROOF AND TERMINATE WITH GOOSENECK. |
| M223 | PROVIDE PVC AIR INTAKE FOR BOILER. INTAKE TO BE SIZED BY BOILER MANUFACTURER. RUN DUCT UP THRU ROOF AND TERMINATE WITH GOOSENECK. |
| M224 | INSULATE OUTSIDE AIR INTAKE. LOCATED INSIDE THE MECHANICAL ROOM, WITH 2" INSULATION. PROVIDE ALUMINUM JACKET. CHEMICAL SHOT FEEDER. |
| M225 | PROVIDE EMERGENCY SHUTOFF SWITCH FOR BOILERS. INSTALL 48" ABOVE FINISHED FLOOR. SEE SHEET M8.4 FOR MORE INFORMATION. |
| M226 | PROVIDE EMERGENCY SHUTOFF SWITCH FOR BOILERS. INSTALL 48" ABOVE FINISHED FLOOR. SEE SHEET M8.4 FOR MORE INFORMATION. |
| M237 | PROVIDE STAINLESS STEEL BOILER EXHAUST SIZED BY BOILER MANUFACTURER. PVC BOILER EXHAUST IS ACCEPTABLE IF ALLOWED BY THE BOILER MANUFACTURER. |
| M238 | BOILER COOLDOWN. SEE DETAIL FOR MORE INFORMATION. |
| M240 | COORDINATE PIPING LOCATION WITH ELECTRICAL CONTRACTOR. TYPICAL THROUGHOUT. |
| M241 | COORDINATE THE DEPTH OF THE CONDENSATE TRAP WITH THE STRUCTURAL CONTRACTOR. |

ISSUED FOR

REV DATE

CONTRACT DOCUMENTS
DESIGN DEVELOPMENT

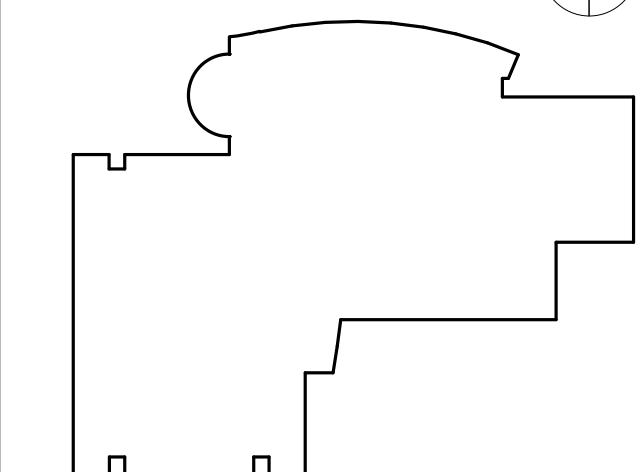
11.04.2019
11.08.2019

SEALS AND SIGNATURES



KEYPLAN

PLAN NORTH



DRAWING TITLE

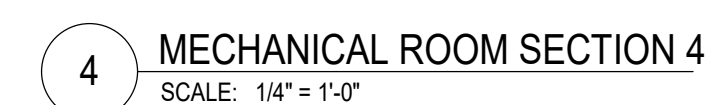
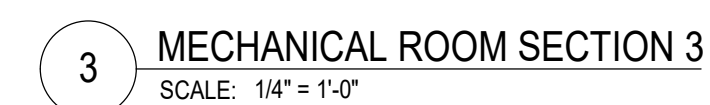
**MECHANICAL ROOM
ENLARGED PIPING
AND DUCT PLANS**

SCALE
1/4" = 1'-0"

PROJECT NUMBER

M4.1

DRAWING NUMBER





1. PROVIDE ONE CHEMICAL SHOT FEEDER FOR THE CHILLED WATER SYSTEM AND ONE CHEMICAL SHOT FEEDER FOR THE HOT WATER SYSTEM.

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M501	PROVIDE HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURN DOWN RATIO OF 300:1 FOR FLOW BYPASS. BASIS OF DESIGN VSI, SERIES V. BUTTERFLY VALVES ARE NOT ACCEPTABLE FOR FLOW BYPASS.
------	---

M502	INSTALL BOILERS ON SPRING TYPE VIBRATION ISOLATOR.
M503	SELECT VALVE FOR A 5PSI DROP AT 20GPM.
M504	SELECT VALVE FOR 5PSI DROP AT MINIMUM CHILLER FLOW AS REQUIRED BY MANUFACTURER.

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CONTRACT DOCUMENTS	01.04.2019
DESIGN DEVELOPMENT	11.08.2018

SEALS AND SIGNATURES



DRAWING TITLE

HOT WATER AND CHILLED WATER PIPING DIAGRAM

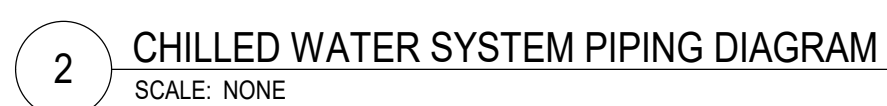
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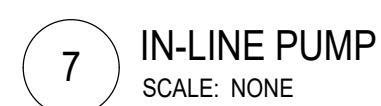
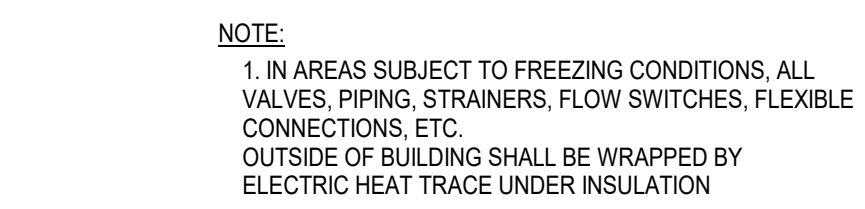
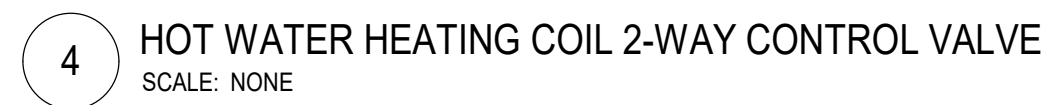
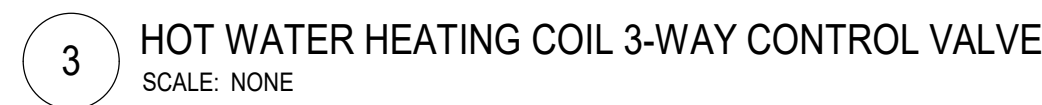
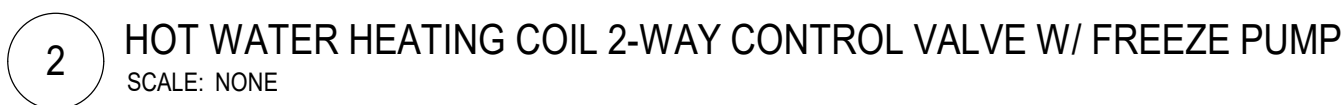
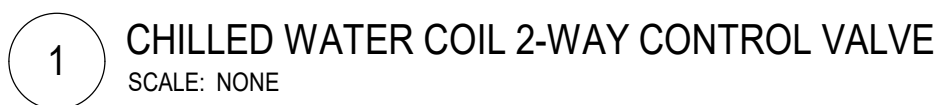
SCALE 10822.000

PROJECT NUMBER

M5.1

DRAWING NUMBER



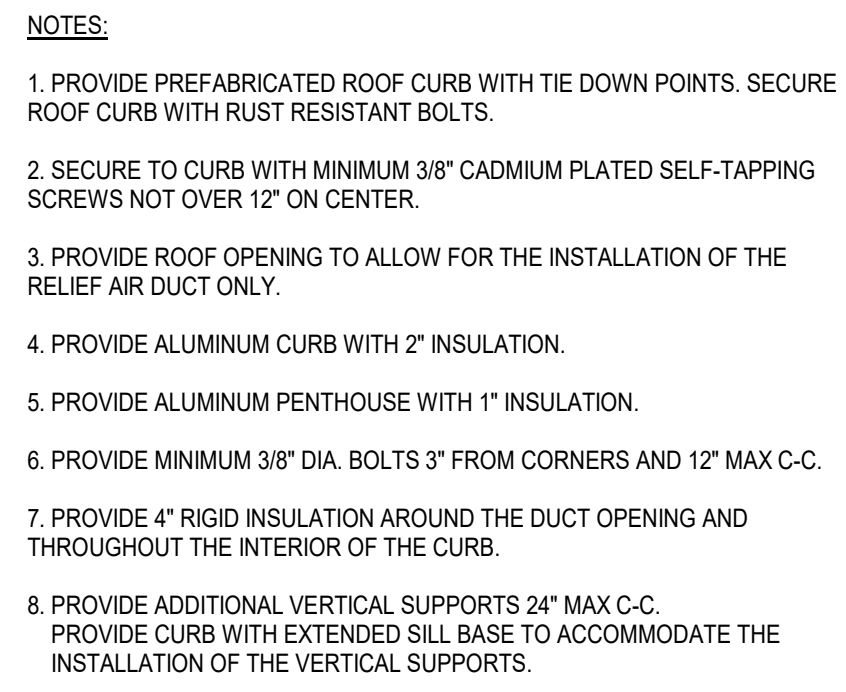
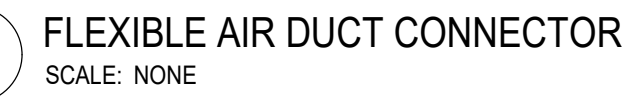




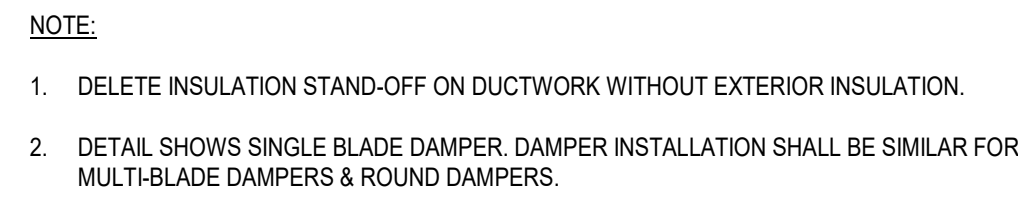
NOTE:

- 1. ALL VANE ELBOWS SHALL BE CONSTRUCTED AND INSTALLED AS DETAILED BY SMACNA.
- 2. WHEN W1 DOES NOT EQUAL W2, VANE SHALL BE SINGLE THICKNESS VANE TYPE REGARDLESS OF W DIMENSION.
- 3. ALL SINGLE THICKNESS VANES SHALL HAVE A 2" [50mm] RADIUS, 1 1/2" [40mm] MAXIMUM SPACE BETWEEN VANES AND A 3/4" [20mm] TRAILING EDGE.
- 4. WHEN W EQUALS W2 AND W1 IS GREATER THAN 20" [500mm] VANES SHALL BE DOUBLE VANE TYPE.

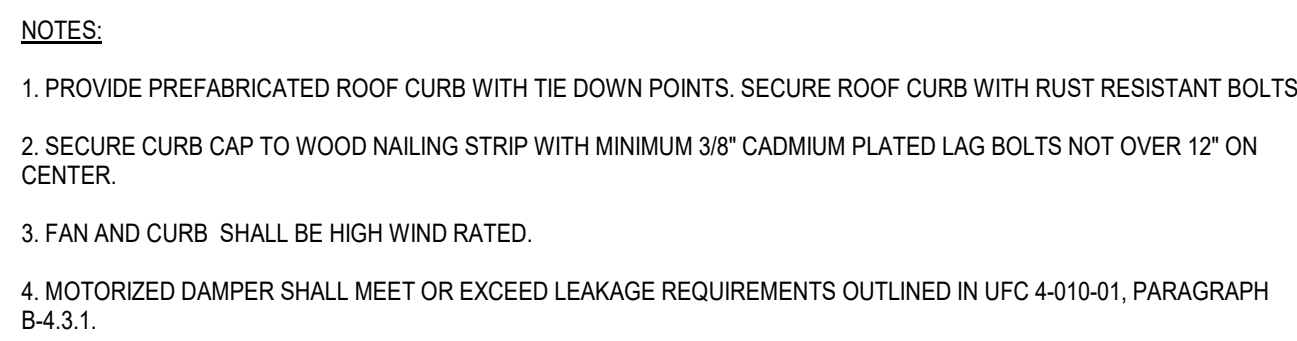
2 DUCTWORK SQUARE VANE ELBOWS
SCALE: NONE



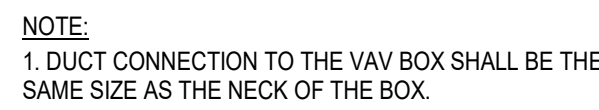
4 ARCHITECTURAL LOUVERED PENTHOUSE
SCALE: NONE



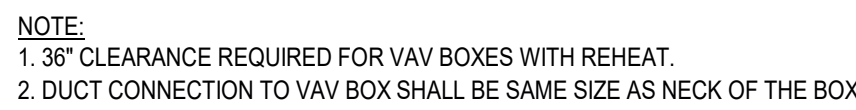
5 VOLUME DAMPER DETAIL
SCALE: NONE



6 BELT DRIVEN POWER ROOF VENTILATOR
SCALE: NONE



8 TYPICAL VAV TERMINAL UNIT CONNECTIONS
SCALE: NONE



9 EXHAUST OR RETURN BRANCH DUCTWORK
SCALE: NONE



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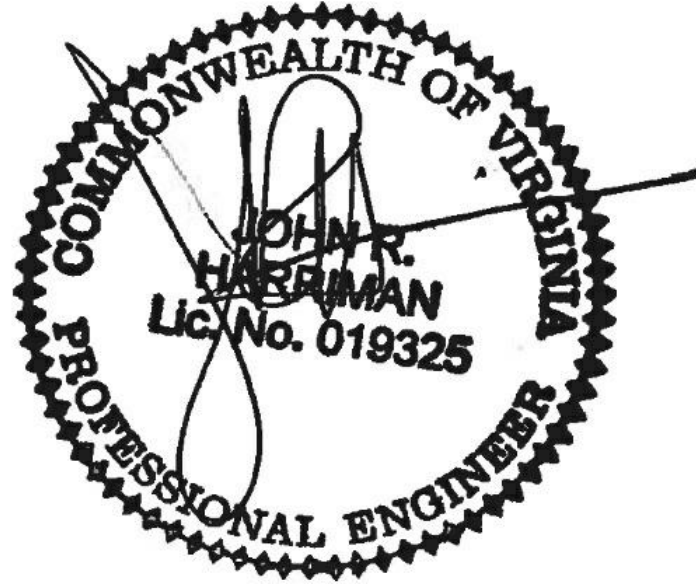
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SEALS AND SIGNATURES



KEYPLAN

DRAWING TITLE

MECHANICAL DETAILS - DUCTWORK

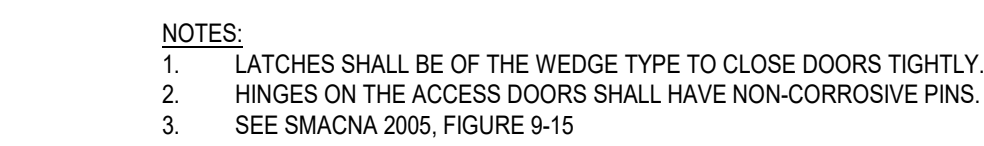
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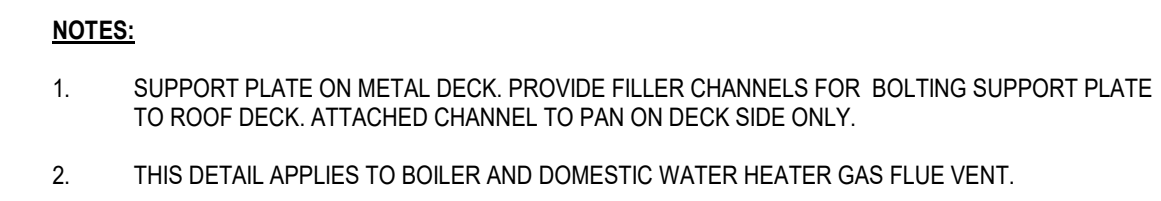
PROJECT NUMBER

M6.2

DRAWING NUMBER



1 ACCESS PANEL AND DOOR DETAIL
SCALE: NONE



3 GAS FLUE VENT
SCALE: NONE

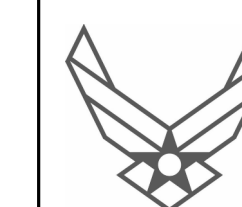


4 CHEMICAL SHOT FEEDER DETAIL
SCALE: NONE

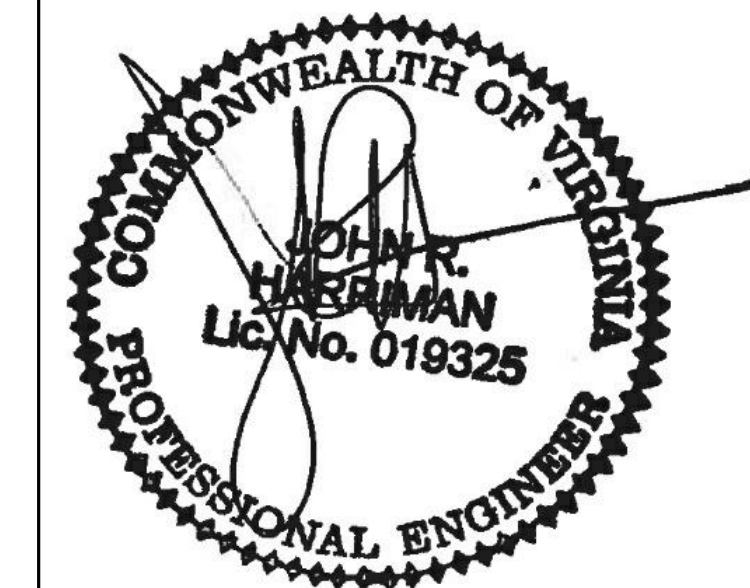


5 CHILLED WATER BUFFER TANK
SCALE: NONE



[illegible]

SEALS AND SIGNATURES



KEYPLAN

DRAWING TITLE

MECHANICAL DETAILS

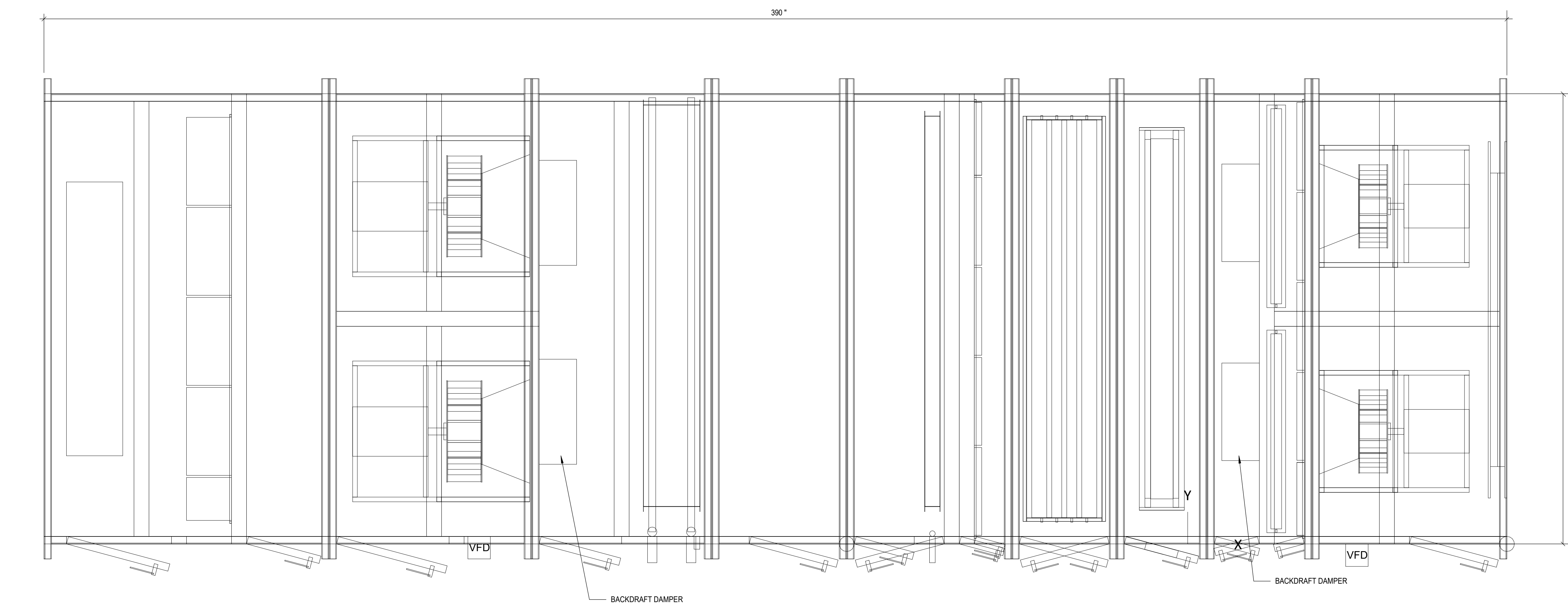
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SCALE

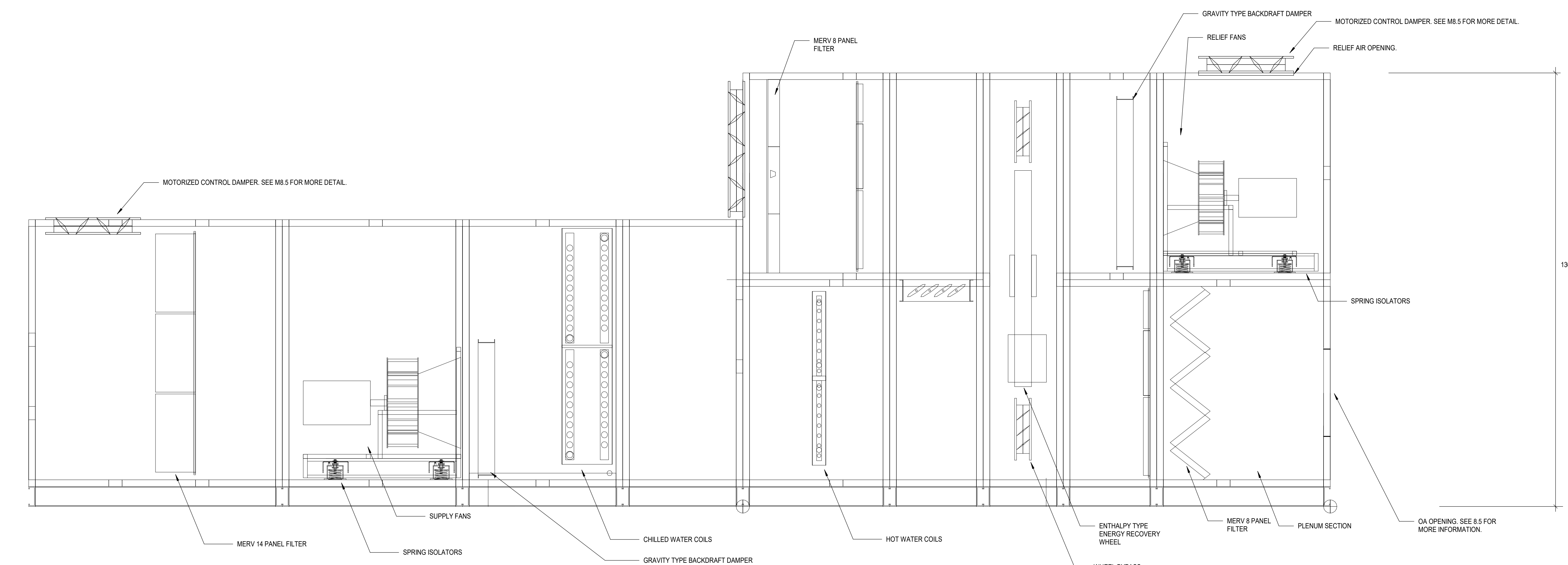
PROJECT NUMBER

M6.4

DRAWING NUMBER



1 AIR HANDLING UNIT PLAN VIEW
SCALE: NONE



2 AIR HANDLING UNIT ELEVATION VIEW
SCALE: NONE

NOTES:			
1.	SEE DETAIL SHEET M6.4 FOR AIR HANDLING UNIT CONFIGURATION.	5.	PROVIDE SOLID PANELS WITH GALVANIZED STEEL INTERIOR LINER.
2.	PROVIDE VFD FOR EACH FAN. PROVIDE HAND OFF TO AUTO SWITCH.	6.	PROVIDE 2" THICK PANELS WITH R13 INJECTED FOAM INSULATION.
3.	PROVIDE MERV 4 PRE-FILTER AND MERV 14 FILTER.	7.	PROVIDE INSULATION WINDOW FOR EACH RECESS SECTION.
4.	PROVIDE DIFFERENTIAL PRESSURE READING PORTS ON EACH ACCESS MODULE.	8.	PROVIDE ALUMINUM FAN BLOWER.
		9.	PROVIDE 12 BLADE FANS WITH 2" SPRING ISOLATOR.
		10.	CASING LEAKAGE SHALL NOT EXCEED ASHRAE 111 CL6 AT DESIGN PRESSURE UP TO POSITIVE OR NEGATIVE 8 I.W.G.
		11.	PROVIDE BACKDRAFT DAMPER FOR EACH RECESS FAN AND SUPPLY AIR FAN.
		12.	PROVIDE LIGHTING PER EACH ACCESS MODULE AND COORDINATE WITH ELECTRICAL.
		13.	PROVIDE ENERGY WHEEL WITH 460V/3PH 60HZ MOTOR AND MOTOR STARTER. COORDINATE WITH ELECTRICAL AND CONTROLS CONTRACTOR.

AIR HANDLING UNIT HYDRONIC COIL SCHEDULE																	
SIGN FLOW F/M	CAPACITY MBH	QUANTITY OF COILS IN BANK	MIN ROWS	MAX FINS PER INCH	MAXIMUM FACE VELOCITY	COIL SIZE DATA		AIRSIDE DATA					WATERSIDE DATA				REMARKS
						LENGTH	HEIGHT	EAT DB °F	EAT WB °F	LAT DB °F	LAT WB °F	MAX AIR PD I.W.G	EWI °F	LWT °F	FLOW GPM	WATER MAX PD FT. H2O	
5000	1267	2	10	9	510	107"	33"	81.00	66.70	49.70	49.50	1.19	44	54	252.20	16.60	
1000	476	2	1	10	606	104"	24"	55.00	0.00	75.70	0.00	0.23	140	109	30.60	10.00	

CHILLER SCHEDULE (ELECTRIC AIR-COOLED)																		
REF DATA			ELECTRICAL								ELECTRICAL			STARTER TYPE	OPERATING WEIGHT LBS	MANUFACTURER	MODEL	REMARKS
FLOW	NUMBER OF PASSES	WATER MAX PD FT.H2O	CONDENSER DATA			COMPRESSOR DATA					MAX KW AT DESIGN	VOLTAGE	PHASE					
			AMBIENT TEMPERATURE °F	FAN QTY	FAN FLA (EA)	COMPRESSOR QTY	COMP 1 RLA	COMP 2 RLA	COMP 3 RLA	COMP 4 RLA								
	2	9.20	92.2	8	3.6	4	37.8	54.5	37.8	54.5	122.5	460	3	ACL	5528.00	DAIKIN	AGZ110E	

NOTES:	
1. PROVIDE THE FOLLOWING:	3. CHILLER SHALL EXCEED ASHRAE 90.1-2010 MINIMUM PERFORMANCE REQUIREMENTS, AS FOLLOWS:
- LOW AMBIENT OPERATION DOWN TO 0°F.	- EER: 9.562 PER AHR 550/350
- VFD FOR THE CONDENSER FANS	- IPLV: 12.75 PER AHR 550/350
- HIGH AMBIENT OPERATION, 125°F	
- FACTORY SUPPLIED SPRING ISOLATORS	4. PROVIDE HARDWARE INTERFACE BETWEEN CHILLER PANEL AND SITE DDC CONTROLS.
- COMPRESSOR SOUND BLANKETS	

NOTES:

1. PROVIDE STAINLESS STEEL EXHAUST AND PVC INLET.
2. SEE SHEET M5.1 FOR HOT WATER PIPING DIAGRAM AND ADDITIONAL HOT WATER SYSTEM COMPONENTS.

NOTES:

1. PROVIDE VFD FOR EACH HW AND CW PUMP.
2. PROVIDE INERTIA PADS FOR EACH HW AND CW PUMP.

NOTES:

1. ALL FANS SHALL BE RATED FOR HIGH WIND.
2. PROVIDE TIE DOWN POINTS
3. PROVIDE FAN WITH SPEED CONTROLLER.
4. FAN ALTERNATIVE DIRECT DRIVE: MODEL TYPE G



GRILLE, REGISTER, AND DIFFUSER SCHEDULE														
MARK	TYPE	MINIMUM AIRFLOW [CFM]	MAXIMUM AIRFLOW [CFM]	NECK SIZE	LINEAR DIFFUSER DATA			FACE SIZE	MAX PD [IN-W.G.]	MAX NC	BASIS OF DESIGN		REMARKS	
					LENGTH [FT]	SLOT QUANTITY	SLOT WIDTH [IN]				MANUFACTURER	MODEL		
S1-1	SQUARE PLAQUE	0	150	6	-	-	-	24x24	0.06	10	TITUS	OMNI		
S1-2		151	200	8	-	-	-		0.06	10				
S1-3		201	320	10	-	-	-		0.06	10				
S1-4		321	390	12	-	-	-		0.06	10				
S3-1	LINEAR SLOT SUPPLY	0	100	6	-	-	-	12x12	0.08	10	TITUS	TBD-30-48-8 TBD-30-60-8		
S2-1		0	200	8	4	3	1		-	0.05				24
S2-2		0	200	8	5	3	3/4"		-	0.05				24
S2-4		0	300	10	6	3	3/4"		-	0.05				20
S2-5	LINEAR SLOT SUPPLY	0	200	8	8	3	3/4"	-	0.05	21	TITUS	ML-38		
S2-6		0	100	6	12	3	3/4"	-	0.05	17				
S2-7		0	200	8	12	3	3/4"	-	0.05	25				
S4-1		0	130	6	-	-	-	6Ø	0.05	10	TITUS	R-OMNI		
R1-1	ROUND PLAQUE	0	130	6	-	-	-		0.08	10	TITUS	OMNI		
R1-2		131	200	8	-	-	-		0.08	10				
R1-3		201	320	10	-	-	-		0.08	10				
R1-4		321	400	12	-	-	-	0.08	10					
R1-5	CEILING RETURN	401	530	14	-	-	-	-	0.08	10	TITUS	350RL		
R3-1		0	1500	-	-	-	-	24x24	0.04	10				
R3-2		0	3050	-	-	-	-	24x48	0.04	10				
R3-3		0	600	-	-	-	-	8x30	0.04	10				
E1-1	RETURN GRILLE	0	100	6	-	-	-	12X12	0.1	10	TITUS	OMNI		
E1-2		101	200	8	-	-	-		0.1	10				
E1-3		0	320	10	-	-	-		24x24	0.1				10
E1-3		0	320	10	-	-	-		24x24	0.1				10

NOTES:
1. PROVIDE: ALL SUPPLY AIR OMNI AIR TERMINALS WITH FACTORY INSTALLED R6 FOIL FACED INSULATION.
2. FIELD INSULATE ALL PLENUM SLOT DIFFUSERS WITH R6 FOIL FACED INSULATION.
3. SIZES OF THE DUCT CONNECTIONS TO SUPPLY AIR, RETURN AIR, AND EXHAUST AIR TERMINALS SHALL MATCH THE NECK SIZE OF THEIR TERMINALS UNLESS OTHERWISE NOTED ON THE DRAWINGS.
4. BASIS OF DESIGN: TITUS

HOT WATER FAN COIL UNIT SCHEDULE																		
TAG	LOCATION	TYPE	AIRSIDE DATA			FAN DATA			HEATING COIL DATA					MAX NC	OPERATING WEIGHT LBS	MANUFACTURER	MODEL	REMARKS
			AIR FLOW CFM	VOLUME CONTROL	EXTERNAL SP I.W.G	WATTS	VOLTAGE	PHASE	EAT DB °F	LAT DB °F	EWI °F	LWT °F	FLOW GPM					
FCU-1	VESTIBULE 100	HORIZONTAL CONCEALED	200	CV	0.50	75 W	120 V	1	55	95	140	110	0.60	5.50	45	120.00	DAIKIN	THC-H02

AIR-COOLED CONDENSING UNIT SCHEDULE																
TAG	LOCATION	TYPE	REFRIGERANT	NOMINAL CAPACITY TONS	AMBIENT TEMP °F	CONDENSER FANS		ELECTRICAL DATA					OPERATING WEIGHT LBS	MANUFACTURER	SYSTEM	REMARKS
						QUANTITY	CONDENSER FAN CFM	MAX KW AT DESIGN	VOLTAGE	PHASE	FLA	MOCP				
CU-1	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	
CU-2	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	
CU-3	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	
CU-4	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	
CU-5	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	
CU-6	ROOF	AIR COOLED	R410A	2	95	1	2101	1.7	208	1	12	20	108.00	DAIKIN	RK24NMVJU	

NOTES:
1. PROVIDE UNIT WITH LOW AMBIENT OPERATION DOWN TO 0°F.
2. PROVIDE UNIT WITH HIGH AMBIENT OPERATION UP TO 115°F

EXPANSION TANK SCHEDULE																
TAG	SYSTEM	LOCATION	TYPE	TEMPERATURE RANGE		TANK VOLUME GAL	MINIMUM ACCEPTANCE VOLUME GAL	SIZE DATA			OPERATING PRESSURE		FULL WEIGHT LBS	MANUFACTURER	MODEL	REMARKS
				MINIMUM °F	MAXIMUM °F			DIAMETER	HEIGHT	CONNECTION SIZE	MIN PSI	MAX PSI				
ET-1	HOT WATER	MECHANICAL ROOM	BLADDER	40	140	80.0	80.0	24"	51"	1"	10.00	100.00	928	Bell & Gossett	B-300	
ET-2	CHILLED WATER	MECHANICAL ROOM	BLADDER	40	95	53.0	53.0	24"	37"	1"	10.00	100.00	629	Bell & Gossett	B-200	

DX FAN COIL UNIT SCHEDULE																			
TAG	LOCATION	SYSTEM	TYPE	AIRSIDE DATA			FAN DATA			COOLING COIL DATA					MAX NC	OPERATING WEIGHT LBS	MANUFACTURER	MODEL	REMARKS
				AIR FLOW CFM	VOLUME CONTROL	EXTERNAL SP I.W.G	WATTS	VOLTAGE	PHASE	COOLING COIL DATA		EAT DB °F	EAT WB °F	LAT DB °F					
										TOTAL CAPACITY BTU/HR	SENSIBLE CAPACITY BTU/HR								
EV-1	ELEC 350	CU-1	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU
EV-2	COMM 348	CU-2	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU
EV-3	COMM 226	CU-3	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU
EV-4	ELEC 111	CU-4	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU
EV-5	STORAGE/KILN 338.1	CU-5	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU
EV-6	STORAGE/KILN 338.1	CU-6	DX-WALL MOUNTED	600	4 SPEED	0.20	0	208	1	21200	15760	75.0	62.6	53.0	52.5	53	27.00	DAIKIN	FTK24NMVJU

UNIT HEATER SCHEDULE (HOT WATER)																				
TAG	LOCATION	TYPE	AIRSIDE DATA					HEATING DATA					FAN DATA				MOUNTING HEIGHT	MANUFACTURER	MODEL	REMARKS
			LOW SPEED CFM	HIGH SPEED CFM	EAT DB °F	LAT DB °F	THROW	MINIMUM CAPACITY BTU/HR	EWI °F	LWT °F	FLOW GPM	MAX PD FT.H2O	HORSEPOWER	RPM	VOLTAGE	PHASE				
UH-1	MECHANICAL ROOM	HORIZONTAL DISCHARGE	950	1100	55	95	29'-0"	48400	140	110	3.20	3.00	0.05	1000	120	1	6'-0"	TRANE	S-72	
UH-2	MECHANICAL ROOM	HORIZONTAL DISCHARGE	950	1100	55	95	29'-0"	48400	140	110	3.20	3.00	0.05	1000	120	1	6'-0"	TRANE	S-72	

SILENCER SCHEDULE																				
TAG	SYSTEM	LOCATION	TYPE	DESIGN AIR FLOW CFM	FACE VELOCITY FPM	MAXIMUM AIR PD I.W.G.	DUCT SIZE		LENGTH	DYNAMIC INSERTION LOSS BY OCTAVE BAND (DB RE 10 ⁴ -(12)) W								MANUFACTURER	MODEL	REMARKS
							WIDTH	HEIGHT		1 63 HZ	2	3	4 500 HZ	5 1,000 HZ	6 2,000 HZ	7 4,000 HZ	8 8,000 HZ			
S-1	AHU-1	MAIN SUPPLY	DISSIPATIVE RETANGULAR	22000	1140	0.25	70"	40"	7'-0"	62	57	50	48	51	52	45	35	VIBRO ACOUSTICS	RD-MV-F4	
S-2	AHU-1	MAIN RETURN	DISSIPATIVE RETANGULAR	18500	1060	0.25	70"	36"	7'-0"	58	55	53	55	57	57	47	34	VIBRO ACOUSTICS	RD-MV-F4	

FIN TUBE RADIATION SCHEDULE																		
TAG	LOCATION	LENGTH	MOUNTING	MINIMUM FINS PER FOOT	ENCLOSURE		CAPACITY PER FT. BTU/HR	MINIMUM CAPACITY BTU/HR	MINIMUM WATER FLOW GPM	EWT °F	LWT °F	EAT DB °F	HORIZONTAL ROWS	VERTICAL ROWS	DRY WEIGHT LBS	MANUFACTURER	MODEL	REMARKS
					WIDTH	HEIGHT												
FTR-1	CENTRAL PARK 103	15'- 0"	WALL	32	0'- 2"	0'- 8 5/8"	400	6000	0.6 GPM	140	110	65	1	3	51.00 lb	RUNTAL	RF-3	FACTORY CURVED
FTR-1B	CENTRAL PARK 103	15'- 0"	WALL	32	0'- 2"	0'- 8 5/8"	400	6000	0.6 GPM	140	110	65	1	3	51.00 lb	RUNTAL	RF-3	FACTORY CURVED
FTR-2	CENTRAL PARK 103	15'- 0"	WALL	32	0'- 2"	0'- 8 5/8"	400	6000	0.6 GPM	140	110	65	1	3	51.00 lb	RUNTAL	RF-3	FACTORY CURVED
FTR-2B	CENTRAL PARK 103	15'- 0"	WALL	32	0'- 2"	0'- 8 5/8"	400	6000	0.6 GPM	140	110	65	1	3	51.00 lb	RUNTAL	RF-3	FACTORY CURVED

NOTES:
1. SPLIT EACH RADIATOR INTO TWO SECTIONS. ONE(1) AT 5'-0" AND ONE(1) AT 10'-0".
2. RADIATOR LENGTH REPRESENTS THE ARC LENGTH.
3. FINAL RADIATOR ARC LENGTHS SHALL BE BASED ON FIELD CONDITIONS TAKE FINAL MEASUREMENTS PRIOR TO ORDERING THE RADIATORS.

ARCHITECTURAL LOUVERED PENTHOUSE SCHEDULE												
TAG	SYSTEM	LOCATION	DESIGN AIR FLOW CFM	MAXIMUM AIR PD I.W.G	ROOF OPENING		PENTHOUSE THROAT SIZE			MANUFACTURER	MODEL	REMARKS
					LENGTH	WIDTH	LENGTH	WIDTH	HEIGHT			
ALP-1	AHU-1	ROOF	25000	0.10	5'-0"	4'-6"	10'-0"	5'-0"	4'-6"	RUSKIN	PHB637HDXD	
ALP-2	AHU-1	ROOF	18500	0.10	5'-0"	4'-2"	10'-0"	5'-0"	4'-6"	RUSKIN	PHB637HDXD	
ALP-3	AHU-1	ROOF	1000	0.10	2'-0"	2'-0"	2'-0"	2'-0"	1'-4"	RUSKIN	PHB637HDXD	

VOLUME CONTROL BOX SCHEDULE															MANUFACTURER	MODEL	REMARKS	
TAG	INLET DIAMETER	AIRSIDE DATA		MAXIMUM NOISE DATA	AIRSIDE DATA		HYDRONIC REHEAT COIL DATA											
		MINIMUM AIR FLOW CFM	MAXIMUM AIR FLOW CFM		RADIATED SOUND (NC)	DISCHARGE SOUND (NC)	SP INLET SF	MAX AIR FLOW G	EAT (°F)	LAT (°F)	EWT (°F)	LWT (°F)	SENSIBLE CAPACITY (BTU/HR)	PIPE FLOW (GPM)	PIPE CONNECT ON SIZE IN	MAX WATER PD FT @ 120		
VAV-1	8"	350	550	30	35	1.00	0.60	55	95	140	110	15400	1.1	3/4"	5.00	PRICE	SDV	
VAV-2	7"	200	400	30	35	1.00	0.60	55	95	140	110	8800	0.6	3/4"	5.00	PRICE	SDV	
VAV-3	10"	600	950	30	35	1.00	0.60	55	95	140	110	26400	1.8	3/4"	5.00	PRICE	SDV	
VAV-4	8"	400	550	30	35	1.00	0.60	55	95	140	110	17600	1.2	3/4"	5.00	PRICE	SDV	
VAV-5	4"	150	150	30	35	1.00	0.60	55	95	140	110	6600	0.5	3/4"	5.00	PRICE	SDV	
VAV-6	4"	100	100	30	35	1.00	0.60	55	95	140	110	4400	0.3	3/4"	5.00	PRICE	SDV	
VAV-7	4"	100	200	30	35	1.00	0.60	55	95	140	110	4400	0.5	3/4"	5.00	PRICE	SDV	
VAV-8	9"	400	700	30	35	1.00	0.60	55	95	140	110	17600	1.2	3/4"	5.00	PRICE	SDV	
VAV-9	300	300	400	30	35	1.00	0.60	55	95	140	110	13200	0.9	3/4"	5.00	PRICE	SDV	
VAV-10	6"	300	300	30	35	1.00	0.60	55	95	140	110	13200	0.9	3/4"	5.00	PRICE	SDV	
VAV-11	6"	300	300	30	35	1.00	0.60	55	95	140	110	13200	0.9	3/4"	5.00	PRICE	SDV	
VAV-12	6"	300	300	30	35	1.00	0.60	55	95	140	110	13200	0.9	3/4"	5.00	PRICE	SDV	
VAV-13	6"	200	300	30	35	1.00	0.60	55	95	140	110	8800	0.6	3/4"	5.00	PRICE	SDV	
VAV-14	4"	200	200	30	35	1.00	0.60	55	95	140	110	8800	0.6	3/4"	5.00	PRICE	SDV	
VAV-15	7"	400	400	30	35	1.00	0.60	55	95	140	110	17600	1.2	3/4"	5.00	PRICE	SDV	
VAV-16	6"	200	300	30	35	1.00	0.60	55	95	140	110	8800	0.6	3/4"	5.00	PRICE	SDV	
VAV-17	6"	200	300	30	35	1.00	0.60	55	95	140	110	8800	0.6	3/4"	5.00	PRICE	SDV	
VAV-18	8"	500	500	30	35	1.00	0.60	55	95	140	110	22000	1.5	3/4"	5.00	PRICE	SDV	
VAV-19	4"	150	150	30	35	1.00	0.60	55	95	140	110	6600	0.5	3/4"	5.00	PRICE	SDV	
VAV-20	6"	300	300	30	35	1.00	0.60	55	95	140	110	13200	0.9	3/4"	5.00	PRICE	SDV	
VAV-21	5"	250	250	30	35	1.00	0.60	55	95	140	110	11000	0.8	3/4"	5.00	PRICE	SDV	
VAV-22	5"	250	250	30	35	1.00	0.60	55	95	140	110	11000	0.8	3/4"	5.00	PRICE	SDV	
VAV-23	5"	250	250	30	35	1.00	0.60	55	95	140	110	11000	0.8	3/4"	5.00	PRICE	SDV	
VAV-24	4"	150	200	30	35	1.00	0.60	55	95	140	110	6600	0.5	3/4"	5.00	PRICE	SDV	
VAV-25	4"	150	150	30	35	1.00	0.60	55	95	140	110	6600	0.5	3/4"	5.00	PRICE	SDV	
VAV-26	8"	350	450	30	35	1.00	0.60	55	95	140	110	15400</						

DRAWING NUMBER



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SEALS AND SIGNATURES



KEYPLAN

DRAWING TITLE

MECHANICAL CONTROLS - LEGEND

NONE

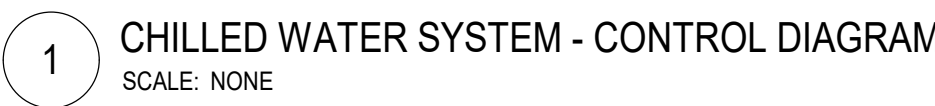
SCALE

PROJECT NUMBER

MO 1

DRAWING NUMBER

M8.1



1. GENERAL:
THE CHILLED WATER SYSTEM SHALL CONSIST OF A SINGLE AIR COOLED CHILLER AND TWO VARIABLE PRIMARY CHILLED WATER PUMPS. THE CHILLED WATER SYSTEM SHALL BE AVAILABLE TO OPERATE 24 HOURS A DAY, 365 DAYS A YEAR. ALL FUNCTIONS AND ALARMS SHALL BE MONITORED BY THE BAS. HARDWARE INTERFACES BETWEEN CHILLER PANEL AND SITE DDC CONTROLS SHALL BE PROVIDED BY THE CHILLER MANUFACTURER. SEQUENCE OF OPERATIONS AND CONTROL OF CHILLER SHALL BE FROM THE CHILLER MANUFACTURER PANEL.

2. CHILLER LOADING/UNLOADING:
THE BAS SHALL CHECK CHILLER CONTROL PANEL ON/UP VERIFICATION AND CONFIRMATION OF PUMP OPERATION FROM THE PUMP DDC CONTROLLER AND WATER FLOW FROM THE WATER FLOW SWITCH. THE CHILLERS PROVIDE MANUFACTURER'S CONTROL PANEL WILL LOAD AND UNLOAD THE CHILLER'S COMPRESSORS TO MAINTAIN THE SUPPLIED CHILLED WATER TEMPERATURE SET POINT SET AT 42°F (ADJUSTABLE).

3. CHILLED WATER PUMP DUTY/STANDBY CONTROL:
THE CHILLED WATER PUMPS ARE CONFIGURED IN A MANNER THAT IF THE OPERATING DUTY PUMP SHOULD FAIL, AS SENSED BY A MISMATCH BETWEEN THE BMS'S SENSED FLOW AND TIME TO FLOW, THE MOST UPSTREAM CURRENT SENSING RELAYS, THE BMS WOULD START THE STANDBY PUMP AND DISABLE THE FAILED PUMP FROM BMS CONTROLLED OPERATION. THE BMS SHALL AUTOMATICALLY ALTERNATE THE OPERATION OF THE DUTY AND STANDBY PUMP ON A LEAD-ARROUND, BASED ON A RUN TIME SCHEDULE. HOS SETTINGS SHALL BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE THROUGH THE DRIVER'S KEYPAD FOR THE WATER PUMP. WHEN THE HAND MODE THE PUMPS SHALL BE STARTED. IN THE OFF MODE THE PUMPS SHALL BE OFF. IN THE AUTO MODE, THE PUMP SHALL BE STARTED AND STOPPED THROUGH THE BMS.

4. CHILLED WATER PUMP SPEED CONTROL:
THE BMS MODULATES THE DUTY CHILLED WATER PUMP TO MAINTAIN DIFFERENTIAL PRESSURE, AS SENSED BY A DIFFERENTIAL PRESSURE SENSOR CONNECTED BETWEEN THE SUPPLY AND RETURN CHILLED WATER PIPING. THE DIFFERENTIAL PRESSURE SENSOR SHALL BE LOCATED IN A LOCATION TO ENSURE DESIGN FLOW AT FURTEST RUNS. THE DIFFERENTIAL PRESSURE SENSOR SHALL BE USED IN MULTIPLE BRANCHES WITH BMS PID CONTROLLING FOR WORST CASE. LOCAL-REMOTE SETTINGS SHALL BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE THROUGH THE DRIVER'S KEYPAD FOR EACH PUMP. IN THE LOCAL MODE, THE PUMP SPEED SHALL BE CONTROLLED THROUGH A MANUAL SPEED CONTROL LOCATED AT THE DRIVER CONTROL PANEL. IN THE REMOTE MODE, THE PUMP SPEED SHALL BE CONTROLLED BY THE BMS.

5. CHILLED WATER BYPASS VALVE CONTROL:
THE CHILLED WATER BYPASS VALVE SHALL BE NORMALLY CLOSED. WHEN THE BMS SENSES THAT THE CHILLER IS RECEIVING INSUFFICIENT FLOW, AS SENSED BY THE CHILLED WATER FLOW METER, THE BMS SHALL MODULATE OPEN THE CHILLED WATER BYPASS VALVE UNTIL SUFFICIENT FLOW IS RESUMED. A DEADBAND PID CONTROL SHALL PROGRAMMED FOR THE BYPASS VALVE CONTROL TO PREVENT SHORT CYCLING OF THE VALVE.

6. FREEZE PROTECTION:
IF THE LEAD CHILLED WATER PUMP IS NOT ALREADY STARTED, THE BMS SHALL TURN ON THE LEAD CHILLED WATER PUMP AND CIRCULATE CHILLED WATER THROUGH THE SYSTEM BASED ON THE FOLLOWING SCHEDULE:
a. IF OUTSIDE AIR TEMPERATURE IS LESS THAN OR EQUAL TO 15°F (ADJUSTABLE), OPERATE THE PUMP CONTINUOUSLY.
b. IF OUTSIDE AIR TEMPERATURE IS BETWEEN 15°F (ADJUSTABLE) AND 25°F (ADJUSTABLE), OPERATE THE PUMP FOR 30 MINUTES EVERY HOUR.
c. IF OUTSIDE AIR TEMPERATURE IS BETWEEN 25°F (ADJUSTABLE) AND 35°F (ADJUSTABLE), OPERATE THE PUMP FOR 10 MINUTES EVERY HOUR.
THE PUMP SHALL OPERATE AT A FREQUENCY OF 15 HZ (ADJUSTABLE) IF NOT REQUIRED OTHERWISE BY THE BMS.
IF THE PUMP IS NOT REQUIRED BY THE AIR HANDLING UNIT, THE CHILLED WATER BYPASS VALVE SHALL MODULATE TO ENSURE THAT THE PUMP CAN OPERATE AS DESCRIBED ABOVE.

7. SET POINT ADJUSTMENT:
ALL SETPOINTS SHALL BE OPERATOR ADJUSTABLE THROUGH THE BMS AT THE CWS.

M501	PROVIDE HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURN DOWN RATIO OF 300:1 FOR FLOW BYPASS. BASIS OF DESIGN VSI, SERIES V. BUTTERFLY VALVES ARE NOT ACCEPTABLE FOR FLOW BYPASS.
M809	PROVIDE INLINE ELECTROMAGNETIC FLOW METER. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.

THE SEQUENCES AND DIAGRAMS OF CONTROL SHOWN ON THIS SHEET CONTAIN THE GENERAL CONTROL REQUIREMENTS. THESE STRATEGIES WILL BE CLARIFIED AND MODIFIED THROUGH PRELIMINARY MEETINGS BETWEEN THE CONTRACTOR, OWNER AND CONSULTANT PRIOR TO IMPLEMENTATION. AT THAT TIME INITIAL SETPOINTS AND RESET SCHEDULES WILL BE DETERMINED. BEFORE THE SYSTEM IS PLACED INTO OPERATION, THE SYSTEM IS OPERATIONAL. TRENDS WILL BE REQUIRED TO VERIFY THE ACCURACY AND ADEQUACY OF THE SEQUENCE OF CONTROL. ADDITIONAL FINE TUNING OR CHANGES TO THE STRATEGY IN OPERATION DURING BUILDING OPERATION WILL BE ACCOMPLISHED BY THE CONTROLS CONTRACTOR AS DIRECTED BY THE OWNER OR ENGINEER. PROVIDE ADDITIONAL ALARMS AS REQUESTED BY THE OWNER OR ENGINEER. ALL SETPOINTS SHALL BE OPERATOR ADJUSTABLE THRU THE BMS AT THE OPERATOR'S WORKING STATION (OWS).



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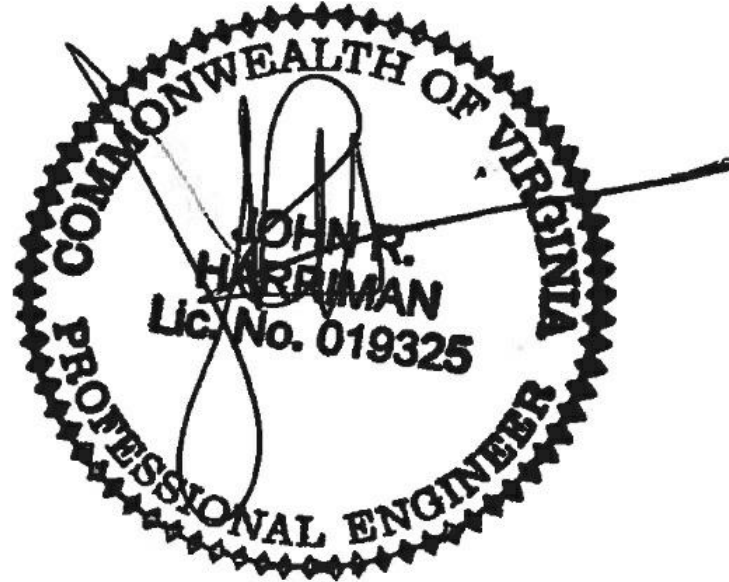
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SEALS AND SIGNATURES



KEYPLAN

DRAWING TITLE

MECHANICAL CONTROLS - CHILLED WATER SYSTEM

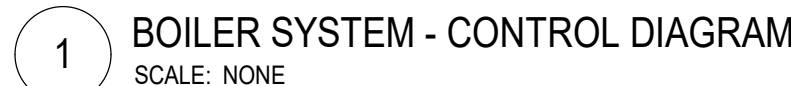
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SCALE 10832 000

PROJECT NUMBER

M8.3

DRAWING NUMBER



1. **GENERAL:**

THE HEATING HOT WATER SYSTEM SHALL CONSIST OF TWO HOT WATER BOILERS, TWO BOILER CONTROL PANELS (ONE PER BOILER), AND TWO VARIABLE PRIMARY HOT WATER PUMPS. THE HEATING HOT WATER SYSTEM SHALL BE AVAILABLE FOR OPERATION 24 HOURS A DAY, 365 DAYS IN A YEAR.

2. BOILER FIRE RATES & STAGING

2. BOILER FIRE RATES & STAGING

THE BMS MONITORS THE OPERATION OF THE BOILERS

THE MASTER BOILER CONTROL PANEL STAGES ON OR OFF THE BOILERS TO MAINTAIN HEATING HOT WATER SUPPLY TEMPERATURE

THE BOILER CONTROL PANEL, ONCE ON, FOR EACH BOILER SHALL CONTROL THE FIRING RATES AS FOLLOWS

- BOILERS 1 AND 2 ARE CONTROLLED IN SEQUENCE TO MAINTAIN HOT WATER SUPPLY TEMPERATURE AT SENSOR T-6.
- IF HOT WATER SUPPLY TEMPERATURE AT T-5 FALLS BELOW ITS SETPOINT OF 150°F (ADJUSTABLE), THE BOILER SEQUENCING SYSTEM STARTS LEAD BOILER AND SETS ITS FIRING RATE AT 100%.
- FIRING RATE IS GRADUALLY INCREASED UNTIL TEMPERATURE IS REACHED AT T-1.
- IF FIRING RATE OF LEAD BOILER REACHES 80% AND TEMPERATURE AT T-1 IS STILL BELOW SETPOINT, THE BMS SHALL START THE LAG BOILER.
- FIRING RATES OF LEAD AND LAG BOILERS ARE RESET TO 25%ADJUSTABLE.
- WHEN BOTH BOILERS ARE OPERATING, THE BOILER SEQUENCING SYSTEM CONTROLS THEM AT THE SAME FIRING RATE.
- SYSTEM MODULATES THEIR FIRING RATES TOGETHER AS NECESSARY BETWEEN 25% AND 100% TO MAINTAIN THE SETPOINT AT T-5.
- IF THE REQUIRED FIRING RATE OF THE COMBINED BOILERS FALLS BELOW 25% (ADJUSTABLE) THE LAG BOILER IS STOPPED AND THE FIRING RATE OF THE LEAD BOILER IS SET TO 40% (ADJUSTABLE), IT THEN OPERATES AS ABOVE.

IF LEAD BOILER FALLS, ALTERNATE BOILER WILL BE AUTOMATICALLY STARTED AND AN ALARM WILL BE ANNUNCIATED AT BMS SYSTEM.

THE MASTER BOILER CONTROL PANELS, ON UPON VERIFICATION AND CONFIRMATION OF PUMP OPERATION FROM THE PUMP DDC CONTROLLER AND WATER FLOW FROM THE CORRESPONDENT WATER FLOW METER.

THE BMS SHALL AUTOMATICALLY ALTERNATE THE OPERATION OF THE BOILERS ON A LEAD-LAG ARRANGEMENT, BASED ON A RUN TIME SCHEDULE.

LEAD BOILER MAY BE MANUALLY ALTERED BY OPERATOR THROUGH SEQUENCING SYSTEM.

MINIMUM AND MAXIMUM HOT WATER FLOW THROUGH EACH BOILER, AS SENSED BY THE CORRESPONDENT WATER FLOW METER SHALL BE ADJUSTABLE THROUGH THE BMS.

IF THE OPERATOR DEPENDS ON THE BOILER TO PROVIDE CIRCULATING FLUID, OPERATION AND CLOSE THE CORRESPONDENT ISOLATION VALVE WHEN A BOILER OFF, A BOILER SHALL NOT BE TURNED ON UNTIL FLOW HAS BEEN ESTABLISHED VIA THE CORRESPONDENT FLOW SWITCH.

3. HEATING HOT WATER PUMP DUTY/STANDBY CONTROL

THE HEATING HOT WATER PUMPS ARE CONFIGURED IN A MANNER THAT IF THE OPERATING DUTY PUMP SHOULD FAIL, AS SENSED BY A MISMATCH BETWEEN THE BMS'S CONTROLLED VALUE AND THE BMS STATUS MONITORED BY ASSOCIATED CURRENT SENSING RELAYS, THE BMS WOULD START THE STANDBY PUMP AND DISABLE THE FAILED PUMP FROM THE BMS CONTROLLED OPERATION.

THE BMS SHALL AUTOMATICALLY ALTERNATE THE OPERATION OF THE DUTY AND STANDBY PUMP ON A LEAD-LAG ARRANGEMENT, BASED ON A RUN TIME SCHEDULE. WHEN SETTING SHALL BE PROVIDED TO THE BMS, THE VARIABLE FREQUENCY DRIVE THROUGH THE DRIVE KEYPAD FOR THE HOT WATER PUMPS SHALL BE SET TO RUN IN THE HAND MODE. THE PUMP(S) SHALL BE STARTED, IN THE OFF MODE THE PUMP(S) SHALL BE OFF. IN THE AUTO MODE, THE PUMP SHALL BE STARTED AND STOPPED THROUGH THE BMS.

4. HEATING HOT WATER PUMP SPEED CONTROL

THE BMS MODULATES THE DUTY HEATING HOT WATER PUMP TO MAINTAIN DIFFERENTIAL PRESSURE, AS SENSED BY A DIFFERENTIAL PRESSURE SENSOR CONNECTED BETWEEN THE SUPPLY AND RETURN HEATING HOT WATER PIPING. THE DIFFERENTIAL PRESSURE SENSOR SHALL BE LOCATED IN A LOCATION TO ENSURE FLOW AT FURTEST RUNS. MULTIPLE DIFFERENTIAL PRESSURE SENSORS SHALL BE USED IN MULTIPLE BRANCHES WITH BMS PID CONTROLLING FOR WORST CASE.

LOCAL-REMOTE SETTINGS SHALL BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE THROUGH THE DRIVE'S KEYPAD FOR EACH PUMP. IN THE LOCAL MODE, THE PUMP SPEED SHALL BE CONTROLLED THROUGH A MANUAL SPEED CONTROL LOCATED AT THE DRIVE CONTROL PANEL. IN THE REMOTE MODE, THE PUMP SPEED SHALL BE CONTROLLED BY THE BMS.

5. **HEATING HOT WATER BYPASS VALVE CONTROL:**

THE HEATING HOT WATER BYPASS VALVE SHALL BE NORMALLY CLOSED. WHEN THE BMS SENSES THAT THE BOILER IS RECEIVING INSUFFICIENT FLOW, AS SENSED BY THE CORRESPONDENT WATER FLOW METER, THE BMS SHALL MODULATE OPEN THE HEATING HOT WATER BYPASS VALVE UNTIL SUFFICIENT FLOW IS DETECTED. A DEAD-BANDPOINT SHALL BE PROGRAMMED FOR THE BYPASS VALVE CONTROL TO PREVENT SHORT CYCLING OF THE VALVE. THE DEFAULT MINIMUM FLOW THROUGH EACH BOILER SHALL BE 20 GPM (ADJUSTABLE).

6. **HEATING HOT WATER SUPPLY TEMPERATURE RESET:**

PROVIDE SETTINGS TO ENABLE/DISABLE HEATING HOT WATER SUPPLY TEMPERATURE RESET CONTROL. SEQUENCE:

IN THE DISABLE MODE, THE HOT WATER SUPPLY TEMPERATURE SETPOINT SHALL BE 150°F, REGARDLESS OF THE OUTSIDE AIR TEMPERATURE.

IN THE ENABLE MODE, THE HOT WATER SUPPLY TEMPERATURE SHALL BE RESET BASED ON OUTDOOR DRY-BULB TEMPERATURE USING THE FOLLOWING SCHEDULE:

- a. 120°F SUPPLY TEMPERATURE AT 65.0°F AND ABOVE
- b. 140°F SUPPLY TEMPERATURE AT 55.0°F AND BELOW
- c. RAMPED LINEARLY BETWEEN 140°F AND 120°F AT TEMPERATURES BETWEEN 55.0°F AND 65.0°F.

7. BOILER ISOLATION VALVE CONTROL:

THE BOILER ISOLATION VALVE IS INTENDED TO SERVE TWO PURPOSES: ISOLATION OF THE BOILER AND BALANCING THE FLOW THROUGH EACH BOILER SUCH THAT WHEN BOTH BOILERS ARE RUNNING, EACH BOILER SEES HALF OF THE FLOW.
THE VALVE SHALL BE CONTROLLED BY THE BMS AND NOT BY THE BOILER.
THE BMS SHALL ONLY RECEIVE A SIGNAL FROM THE BOILER CONTROL PANEL TO OPEN THE VALVE.
THE CONTROL SEQUENCE SHALL BE AS FOLLOWS:

- a. THE BOILER CONTROL PANEL SENDS A SIGNAL TO OPEN THE VALVE.
- b. IF ONLY ONE BOILER IS ON, THE ISOLATION VALVE SHALL BE FULLY OPEN.
- c. IF BOTH BOILERS ARE ON, THEN EACH ISOLATION VALVE SHALL START MODULATING TO MAINTAIN EQUAL FLOW THROUGH EACH BOILER.

COORDINATE WITH THE BOILER MANUFACTURER BEFORE IMPLEMENTING THE ISOLATION VALVE CONTROL SEQUENCE.

8. ALL POINTS SHALL BE OPERATOR ADJUSTABLE THROUGH THE BMS.

9. **HARDWIRED EMERGENCY SHUTDOWN SWITCHES:**
PROVIDE EMERGENCY SHUTDOWN SWITCHES TO MANUALLY SHUTDOWN AND DISABLE BOILERS FROM OPERATION IN EMERGENCY CONDITIONS, IN COMPLIANCE WITH ASME CSD-1-2009.

9. HARDWIRED EMERGENCY SHUTDOWN SWITCHES:

PROVIDE EMERGENCY SHUTDOWN SWITCHES TO IMMEDIATELY SHUTDOWN AND DISCONNECT BOILERS FROM OPERATION IN EMERGENCY CONDITIONS, IN COMPLIANCE WITH ASME CSD-1-2009.

[illegible]

1. THE SEQUENCES AND DIAGRAMS OF CONTROL SHOWN ON THIS SHEET CONTAIN THE GENERAL CONTROL REQUIREMENTS. THESE STRATEGIES WILL BE CLARIFIED AND MODIFIED THROUGH MEETINGS BETWEEN THE CONTRACTOR, OWNER AND CONSULTANT PRIOR TO IMPLEMENTATION. AT THAT TIME INITIAL SETPOINTS AND RESET SCHEDULES WILL BE ESTABLISHED BEFORE PRODUCTION. WHEN THE SYSTEM IS OPERATIONAL, TRENDING WILL BE REQUIRED TO VERIFY THE ACCURACY AND ADEQUACY OF THE SYSTEMS OF CONTROL. ADDITIONAL FINE TUNING OR CHANGES MAY BE REQUIRED. THE SEQUENCE OF OPERATION OF THE BUILDING OPERATION WILL BE ACCOMPLISHED BY THE CONTROLS CONTRACTOR AS DIRECTED BY THE OWNER OR ENGINEER. PROVIDE ADDITIONAL ALARMS AS REQUESTED BY THE OWNER OR ENGINEER. ALL ALARMS SHALL BE MONITORED AVAILABLE THROUGH THE BMS AT THE OPERATOR'S WORKING STATION (OWS).



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M501	PROVIDE HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURN DOWN RATIO OF 300:1 FOR FLOW BYPASS. BASIS OF DESIGN VSI, SERIES V. BUTTERFLY VALVES ARE NOT ACCEPTABLE FOR FLOW BYPASS.
M809	PROVIDE INLINE ELECTROMAGNETIC FLOW METER. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.

[illegible]

SEALS AND SIGNATURES



KEYPLAN

DRAWING TITLE

MECHANICAL CONTROLS - HEATING HOT WATER SYSTEM

NONE

SCALE

PROJECT NUMBER

MO 4

DRAWING NUMBER

M8.4

AHU-1 SEQUENCE OF OPERATIONS	
GENERAL 1. THE RELIEF AIR FANS ARE INTENDED TO OPERATE INDEPENDENT OF EACH OTHER SUCH THAT ONE FAN CAN BE OFF WHILE OTHER FAN CAN BE ON. 2. THE SUPPLY AIR FANS ARE INTENDED TO OPERATE INDEPENDENT OF EACH OTHER SUCH THAT ONE FAN CAN BE OFF WHILE THE OTHER FAN CAN BE ON. 3. THE ENTHALPY WHEEL BYPASS DAMPERS SHALL BE NORMALLY CLOSED AND SHALL STAY FULLY OPEN AS LONG AS THE ENTHALPY WHEEL IS NOT COMMANDED ON. SUPPLY FANS AND RELIEF FANS HAND-OFF-AUTO (HOA) OPERATION 1. HOA SETTINGS SHALL BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE THROUGH THE DRIVE'S KEYPAD FOR THE SUPPLY FANS AND RELIEF FANS. 2. ON A COMMAND TO START, WHEN IN THE AUTO OR HAND MODE, THE SUPPLY AIR, RETURN AIR AND RELIEF AIR DAMPERS SHALL OPEN THROUGH A HARDWIRED INTERLOCK WITH THE VFD DAMPER CONTROL. ONCE THE SUPPLY AIR FANS HAVE STARTED, THE SUPPLY AIR FANS SHALL BE WIRED TO THE VFD DAMPER RUN PERMISSIVE CIRCUIT. 3. IN THE OFF MODE, THE OUTSIDE AIR DAMPER, ECONOMIZER DAMPER, RELIEF AIR DAMPER, RETURN AIR DAMPER AND SUPPLY AIR DAMPER SHALL CLOSE AND THE SUPPLY FANS SHALL BE OFF. 4. IN THE AUTO MODE, THE SUPPLY FANS AND RELIEF FANS SHALL BE STARTED AND STOPPED THROUGH THE BMS. ALL DAMPERS SHALL BE OPENED AND CLOSED THROUGH THE BMS. 5. UPON ACTIVATION, SAFETIES SHALL STOP THE SUPPLY FANS IN THE AUTO OR HAND MODES. SUPPLY FAN AND RELIEF FAN LOCAL-REMOTE SPEED CONTROL 1. LOCAL-REMOTE SETTINGS SHALL BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE (VFD) THROUGH THE DRIVE'S KEYPAD FOR EACH FAN. 2. IN THE LOCAL MODE, THE FAN SPEED SHALL BE CONTROLLED THROUGH A MANUAL SPEED CONTROL LOCATED AT THE DRIVE CONTROL PANEL. 3. IN THE REMOTE MODE, THE FAN SPEED SHALL BE CONTROLLED BY THE BMS. RUN CONDITIONS 1. UNIT SHALL BE AUTOMATICALLY OR MANUALLY ENABLED TO RUN IN OCCUPIED OR UNOCCUPIED MODE AS FOLLOWS: a. AUTOMATIC OPERATION IN OCCUPIED/UNOCCUPIED MODES SHALL BE AS DEFINED BELOW IN THE VAV ZONE TERMINAL UNIT INTERFACE, FAN CONTROL, AND OUTDOOR AIR CONTROL SECTIONS. b. OR MANUALLY SELECTED BY USER FROM GRAPHIC INTERFACE. c. WHEN THE UNIT IS STOPPED MANUALLY OR FROM SAFETY FUNCTIONS, THE FANS SHALL BE DE-ENERGIZED. THE MINIMUM AND ECONOMIZER OUTSIDE AIR (OA) DAMPERS SHALL CLOSE. THE RELIEF AIR (RA) DAMPER SHALL CLOSE AND RETURN AIR (RA) DAMPERS SHALL OPEN. THE CHW VALVE SHALL CLOSE COMPLETELY (EXCEPT AS DESCRIBED ELSEWHERE DURING FREEZE/STAT ACTIVATION) AND THE PREHEAT VALVE SHALL BE CONTROLLED AS DESCRIBED IN PREHEAT SECTION. 2. AHU OPTIMAL START a. AN OPTIMAL START ROUTINE SHALL BE UTILIZED. b. AT THE SCHEDULED OCCUPANCY TIME, THE UNIT SHALL START IF NOT ALREADY STARTED BY THE OPTIMAL START ROUTINE, AND THE UNIT SHALL BE SWITCHED TO THE OCCUPIED MODE. 3. SAFETY SHUTDOWNS a. FREEZE PROTECTION: THE UNIT SHALL SHUT DOWN AS DESCRIBED ABOVE AND GENERATE AN ALARM UPON RECEIVING A FREEZE/STAT STATUS ON LEAVING SIDE OF HEATING COIL. MANUAL RESET REQUIRED. b. HIGH AND LOW STATIC SHUTDOWN: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING EITHER A HARD-WIRED HIGH OR LOW STATIC SHUTDOWN SIGNAL. FINAL SETTING SHALL BE ADJUSTED IN THE FIELD, BASED ON APPROXIMATELY 30% SAFETY FACTOR BEYOND NORMAL MAXIMUM OPERATING CONDITIONS TO AVOID NUISANCE TRIPS, BUT NOT TOO EXCEED DUCT CONSTRUCTION PRESSURE CLASS FOR EACH SYSTEM. c. HIGH STATIC SHUTDOWN LOCATED AT SUPPLY FAN DISCHARGE i. LOW STATIC SHUTDOWN LOCATED AT RETURN FAN SUCTION. c. RETURN AIR SMOKE DETECTION: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A RETURN AIR SMOKE DETECTOR STATUS FROM ADDRESSABLE RELAY. d. SUPPLY AIR SMOKE DETECTION: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A SUPPLY AIR SMOKE DETECTOR STATUS FROM ADDRESSABLE RELAY. e. ISOLATION DAMPER END SWITCH: THE UNIT SHALL NOT START IF THE ISOLATION DAMPER IS NOT PROVEN OPEN VIA ITS END SWITCH. THE BMS WILL ALSO MONITOR THE END SWITCH AND GENERATE AN ALARM IF THE DAMPER IS NOT PROVEN OPEN AFTER THE FAN IS COMMANDED TO START. f. THE SEQUENCE APPLIES TO THE SUPPLY AIR ISOLATION DAMPER, RETURN AIR ISOLATION DAMPER AND RELIEF AIR ISOLATION DAMPER. 4. FAN STATUS MONITORING AND ALARMS SHALL BE PROVIDED AS FOLLOWS: a. FAN FAILURE (SUPPLY, RELIEF): COMMANDED ON, BUT THE STATUS IS OFF. b. FAN IN HAND (SUPPLY, RELIEF): COMMANDED OFF, BUT THE STATUS IS ON. c. FAN VFD FAULT (SUPPLY) d. FAN RUNTIME EXCEEDED (SUPPLY, RELIEF): STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.). THE CONTROLLER SHALL HAVE THIS LOGIC INCLUDED, BUT THE DEFAULT SETTING SHALL BE DISABLED. IT SHALL BE ENABLED ONLY BY USER WHEN NEEDED. ZONE VAV AIR TERMINAL UNIT INTERFACE 1. AT A MINIMUM, ALL VAV TERMINAL UNITS SERVED BY THE AHU SHALL BE LINKED WITH ASSOCIATED VAV AHU CONTROLLER TO PERFORM THE FOLLOWING FUNCTIONS. a. ZONE OCCUPANCY SCHEDULE (USER DEFINED FROM GRAPHIC INTERFACE) SHALL NORMALLY AUTOMATICALLY SELECT THE OCCUPIED OR UNOCCUPIED OPERATING MODE OF THE AIR HANDLING UNIT. i. ACTIVATION OF TIME OVERRIDE SWITCH ON ZONE THERMOSTATS SHALL ONLY RESET ZONE HEATING AND COOLING SETPOINTS TO "OCCUPIED" VALUES, BUT SHALL NOT AFFECT OTHER SCHEDULED UNOCCUPIED OPERATING MODE OF AIR HANDLING UNIT. b. DUCT STATIC PRESSURE RESET AS DESCRIBED IN FAN CONTROL SECTION. c. DISCHARGE AIR TEMPERATURE SETPOINT - OPTIMIZED AS DESCRIBED IN THE DISCHARGE TEMPERATURE CONTROL SECTION. FAN CONTROL 1. GENERAL: a. THE SUPPLY FAN SPEED SHALL BE MODULATED TO MAINTAIN THE DUCT STATIC PRESSURE SETPOINT WHICH SHALL BE AUTOMATICALLY RESET TO MEET ZONE AIRFLOW DEMANDS. b. THE OUTSIDE AIR, RETURN AND RELIEF DAMPERS ALSO MUST BE INTEGRATED WITH THE OPERATION OF THE SUPPLY FANS AND RELIEF FANS TO ENSURE MINIMUM VENTILATION AIR AND TO MAINTAIN PROPER PRESSURE RELATIONSHIPS WITHIN THE SYSTEM. c. THE DAMPER CONTROLS ARE DISCUSSED IN THE MINIMUM OA AND ECONOMIZER CONTROL SECTION. 2. OCCUPIED MODE: a. SUPPLY AIR FANS: i. SHALL BE ENABLED TO RUN CONTINUOUSLY, UNLESS SHUTDOWN ON SAFETIES. ii. SUPPLY FAN SPEED CONTROL VIA AIR DUCT STATIC PRESSURE CONTROL - OPTIMIZED: THE CONTROLLER SHALL MEASURE DUCT STATIC PRESSURE AND MODULATE THE SUPPLY FAN VFD SPEED TO MAINTAIN AN OPTIMIZED DUCT STATIC PRESSURE SETPOINT. iii. a) NO SUPPLY FAN SHALL BE ALLOWED TO RUN AT A SUPPLY AIRFLOW OF 2,500 CFM (ADJ.) OR LESS. b) IF THE BMS IS DECREASING THE SPEED OF THE SUPPLY FANS TO MAINTAIN DUCT STATIC PRESSURE SETPOINTS, AND BY DOING THAT, THE AIRFLOW THROUGH EACH SUPPLY FAN IS APPROACHING 2,500 CFM (ADJ.), THE BMS SHALL DISABLE ONE OF THE SUPPLY FANS AND SHALL INCREASE THE SPEED OF THE REMAINING SUPPLY FAN TO MAINTAIN DUCT STATIC PRESSURE SETPOINTS. 1) IF ONLY ONE SUPPLY AIR FAN IS OPERATIONAL AND IF THE BMS IS INCREASING THE SPEED OF THE SUPPLY FAN TO MAINTAIN DUCT STATIC PRESSURE SETPOINTS, AND BY DOING THAT, THE AIRFLOW THROUGH THE SUPPLY FAN IS APPROACHING 6,000 CFM (ADJ.), THEN THE BMS SHALL ENABLE THE SECOND SUPPLY FAN AND SHALL MODULATE THE SPEED OF BOTH SUPPLY AIR FANS TO MAINTAIN DUCT STATIC PRESSURE SETPOINTS. c) AIRFLOW MEASURING STATIONS SHALL CONTINUOUSLY MONITOR THE TOTAL SUPPLY AIR FAN VOLUME AT EACH SUPPLY FAN IN CFM AND SEND OUTPUT TO THE BMS. b. RELIEF AIR FANS: i. THE BMS SHALL SPACE ON/OFF AND SHALL MODULATE THE RELIEF AIR FANS TO MAINTAIN A CONSTANT BUILDING PRESSURE IN ALL OCCUPIED MODES. ii. PROVIDE MINIMUM THREE SPACE PRESSURE SENSORS, COORDINATE THE FINAL LOCATION OF THE SPACE PRESSURE SENSORS WITH THE DESIGNER OF RECORD. iii. THE SPACE PRESSURE SETPOINT SHALL BE 0.025 IW.C (ADJUSTABLE). iv. THE BMS SHALL MODULATE THE SPEED OF THE RELIEF AIR FANS SUCH THAT THE DIFFERENTIAL PRESSURE SETPOINTS ARE MAINTAINED FOR ALL SPACE DIFFERENTIAL PRESSURE SENSORS. v. IF THE BMS IS DECREASING THE SPEED OF THE RELIEF AIR FANS TO MAINTAIN THE SPACE DIFFERENTIAL PRESSURE SETPOINTS, AND BY DOING THAT, THE AIRFLOW THROUGH EACH RELIEF FAN IS GOING BELOW 1,000 CFM (ADJ.), THE BMS SHALL DISABLE ONE OF THE RELIEF FANS AND SHALL INCREASE THE SPEED OF THE REMAINING RELIEF FAN TO MAINTAIN THE SPACE DIFFERENTIAL PRESSURE SETPOINTS. 1) IF ONLY ONE RELIEF AIR FAN IS OPERATIONAL AND IF THE BMS IS INCREASING THE SPEED OF THE RELIEF FAN TO MAINTAIN THE SPACE DIFFERENTIAL PRESSURE SETPOINTS, AND BY DOING THAT, THE AIRFLOW THROUGH THE RELIEF FAN IS REACHING 5,000 CFM (ADJ.), THEN THE BMS SHALL ENABLE THE SECOND RELIEF FAN AND SHALL MODULATE THE SPEED OF BOTH RELIEF AIR FANS TO MAINTAIN THE SPACE DIFFERENTIAL PRESSURE SETPOINTS. vi. IF THE TOTAL RELIEF AIR FLOW, AS MEASURED BY THE RELIEF AIR FANS, NEEDS TO DROP BELOW 2,500 CFM (ADJUSTABLE) TO MAINTAIN THE BUILDING PRESSURE, THE BMS SHALL MODULATE OPEN THE MINIMUM OUTSIDE AIR DAMPER IN ORDER TO MAINTAIN THE SPACE DIFFERENTIAL PRESSURE SETPOINTS. 1) THE RELIEF AIR FLOW SHALL NOT BE ALLOWED TO DROP BELOW 2,000 CFM (ADJUSTABLE), AT EACH RELIEF AIR FAN. IF THE TOTAL RELIEF AIR FLOW IS 2,000 CFM (ADJUSTABLE) AND THE OUTSIDE AIR FLOW AT THE MINIMUM OUTSIDE AIR DAMPER IS 1,000 CFM (ADJUSTABLE) MORE THAN THE DESIGN AIR FLOW, AN ALARM SHALL BE SENT AT THE BMS. 3. UNOCCUPIED MODE: a. THE BMS SHALL START THE UNIT AND OPERATE THE SYSTEM IN THE UNOCCUPIED HEATING MODE IF THE TEMPERATURE IN THREE ZONES (ADJ.) OR MORE, AS SENSED BY THE ASSOCIATED VAV BOX SPACE TEMPERATURE SENSORS, DROP BELOW 55°F (ADJUSTABLE). THE UNIT SHALL OPERATE UNTIL ALL ZONE THERMOSTATS HAVE REACHED THEIR UNOCCUPIED MODE HEATING TEMPERATURE SETPOINT. i. UNOCCUPIED HEATING MODE: 1) THE CONTROL VALVE SERVING THE AHU PREHEAT COIL SHALL MODULATE AS REQUIRED TO MAINTAIN AN AHU DISCHARGE AIR TEMPERATURE OF 70°F (ADJUSTABLE). 2) THE CONTROL VALVES SERVING THE VAV TERMINAL UNITS SHALL MODULATE AS REQUIRED TO MAINTAIN THE UNOCCUPIED MODE HEATING SPACE TEMPERATURE SETPOINTS. b. THE BMS SHALL START THE UNIT AND OPERATE THE SYSTEM IN THE UNOCCUPIED COOLING MODE IF THE TEMPERATURE IN THREE ZONES (ADJ.) OR MORE, AS SENSED BY THE ASSOCIATED VAV BOX SPACE TEMPERATURE SENSORS RISES ABOVE 85°F (ADJUSTABLE). THE UNIT SHALL OPERATE UNTIL ALL ZONE THERMOSTATS HAVE REACHED THEIR UNOCCUPIED MODE COOLING TEMPERATURE SETPOINT. i. UNOCCUPIED COOLING MODE: 1) THE CONTROL VALVE SERVING THE AHU COOLING COIL SHALL MODULATE OPEN TO MAINTAIN AN AHU DISCHARGE AIR TEMPERATURE OF 65°F (ADJUSTABLE). 2) THE CONTROL VALVES SERVING THE VAV TERMINAL UNITS SHALL MODULATE AS REQUIRED TO MAINTAIN THE UNOCCUPIED MODE SPACE COOLING TEMPERATURE SETPOINTS. c. DURING UNOCCUPIED MODE THE OUTSIDER AIR, ECONOMIZER AND RELIEF AIR DAMPERS SHALL BE FULLY CLOSED. d. DURING UNOCCUPIED MODE THE RELIEF AIR FANS SHALL BE DISABLED. e. DURING UNOCCUPIED MODE, WHEN THE UNIT IS EITHER IN COOLING OR HEATING MODE, THE SUPPLY FANS SHALL OPERATE AS DESCRIBED IN THEIR RELATED OCCUPIED MODE SEQUENCE OF OPERATION. 4. FAN SPEED CONTROL AND PRESSURE MONITORING ALARMS SHALL BE PROVIDED AS FOLLOWS: a. HIGH SUPPLY AIR STATIC PRESSURE: IF THE SUPPLY AIR STATIC PRESSURE IS 25% (ADJ.) GREATER THAN MAXIMUM LIMIT OF THE DP SETPOINT. b. LOW SUPPLY AIR STATIC PRESSURE: IF THE SUPPLY AIR STATIC PRESSURE IS 25% (ADJ.) LESS THAN EFFECTIVE RESET SETPOINT. 5. THE CONTROL BANDS, SETPOINT INCREMENT VALUES, SETPOINT DECREMENT VALUES AND ADJUSTMENT FREQUENCIES SHALL BE ADJUSTED AND TUNED TO MAINTAIN MAXIMUM STATIC PRESSURE OPTIMIZATION WITH STABLE SYSTEM CONTROL AND MAXIMUM COMFORT CONTROL. SUPPLY DUCT STATIC PRESSURE SET POINT CONTROL 1. THE BMS SHALL MEASURE DUCT STATIC PRESSURES, AS SENSED BY THE ASSOCIATED DUCT STATIC PRESSURE SENSORS, AND MODULATE THE SUPPLY FAN VFD SPEED TO MAINTAIN AN OPTIMIZED DUCT STATIC PRESSURE SETPOINT AS FOLLOWS: a. EACH AHU CONTROLLER SHALL BE NETWORKED WITH ALL ASSOCIATED VAV TERMINAL UNITS TO OBTAIN AIRFLOW REQUESTS. THE INITIAL DUCT STATIC PRESSURE SETPOINT SHALL BE DETERMINED BY THE TAB CONTRACTOR. b. THE STATIC PRESSURE SETPOINT SHALL BE RESET BASED ON AIRFLOW REQUESTS, DERIVED FROM VAV BOX DAMPER POSITION AND MEETING THE SUPPLY AIRFLOW REQUIREMENTS AT EACH VAV BOX. c. WHEN THERE ARE NO OR LOW SUPPLY AIR FLOW REQUESTS (WHEN ALL VAV BOX DAMPERS ARE THROTTLED CLOSED - ABOVE 90% CLOSED) THE DUCT STATIC PRESSURE SETPOINT SHALL BE INCREMENTALLY RESET DOWN BY 0.1 IW.C (ADJ.) AT A FREQUENCY OF 5 MINUTES (ADJ.) TO A MINIMUM OF 0.3 IW.C (ADJ.) OR THE SUPPLY FAN VFD HAS REACHED ITS LOWEST OPERATING SPEED LIMIT. d. AS AIRFLOW REQUESTS INCREASE, WHEN AT LEAST ONE ZONE DAMPER IS GREATER THAN 99% OPEN, THE REVERSE SHALL OCCUR AND THE DUCT STATIC PRESSURE SETPOINT SHALL INCREMENTALLY RESET UP AT SAME RATE AS ABOVE TO A MAXIMUM OF 1.25 IN H2O (ADJ.). MINIMUM OUTSIDE AIR 1. GENERAL a. WHEN IN OCCUPIED MODE, THE MINIMUM OUTSIDE AIR SHALL BE MEASURED AND THE MINIMUM OUTSIDE AIR DAMPER SHALL BE CONTROLLED TO MAINTAIN VENTILATION AND PRESSURIZATION REQUIREMENTS. b. PROVIDE SEPARATE MODULATING ACTUATORS AND ANALOG I/O FOR MINIMUM OA DAMPER, ECONOMIZER DAMPER, RETURN DAMPER, AND RELIEF DAMPER. c. AN AIRFLOW MEASURING STATION SHALL BE PROVIDED TO CONTINUOUSLY MONITOR THE MINIMUM OUTSIDE AIR VOLUME IN CFM AND SEND OUTPUT TO THE BMS TO MAINTAIN OA CFM SETPOINT AS SYSTEM FANS MODULATE. OA CFM INPUT AND OUTPUT AND OA DAMPER POSITION FEEDBACK SHALL BE MONITORED AND GRAPHICALLY DISPLAYED IN THE BMS. 2. STARTUP SEQUENCE a. UPON STARTUP, OPEN THE RETURN DAMPER 100% AND THE MINIMUM OA DAMPER TO ABSOLUTE MINIMUM POSITION. b. MODULATE MINIMUM OA DAMPER UP TO 100% OPEN IN ATTEMPT TO ACHIEVE MEASURED OUTSIDE AIRFLOW SETPOINT. IF SETPOINT IS NOT ACHIEVED THROUGH THIS ACTION ALONE, THEN MODULATE RETURN AIR DAMPER CLOSED AS NEEDED TO ACHIEVE OUTSIDE AIRFLOW SETPOINT. c. WHEN THE UNIT IS STARTED IN OCCUPIED MODE AND OUTSIDE AIR TEMPERATURE IS LESS THAN 40°F (ADJ.), THE OUTSIDE AIR (OA) DAMPER SHALL RAMP OPEN TO ITS MINIMUM SETPOINT OVER A 5 MINUTE (ADJ.) PERIOD. WHEN OAT IS >40°F, THERE SHALL NOT BE A DELAYED RAMP OPEN PERIOD. d. WHEN SWITCHING FROM UNOCCUPIED TO OCCUPIED MODE, THE OUTSIDE AIR DAMPER SHALL INITIALLY OPEN TO ABSOLUTE MINIMUM OA CFM SETPOINT. THE FAILSAFE OPERATING CONDITION FOR THE DEMAND BASED VENTILATION SEQUENCE SHALL MAINTAIN NORMAL DESIGN MINIMUM OA CFM SETPOINT AS SCHEDULED. 3. RELIEF FAN SYSTEM - INTEGRATED FAN AND DAMPER CONTROL a. ENERGIZE THE SUPPLY AND RELIEF FANS IN UNISON. CONTROL SUPPLY FAN TO MAINTAIN DUCT STATIC PRESSURE SETPOINT. b. ENABLE RELIEF FAN(S) TO MAINTAIN EACH OF THE SPACE DIFFERENTIAL PRESSURE SETPOINTS. c. MODULATE MINIMUM OA DAMPER UP TO 100% OPEN IN ATTEMPT TO ACHIEVE MEASURED OUTSIDE AIRFLOW SETPOINT. d. IF MINIMUM OA SETPOINT CANNOT BE ACHIEVED WITH MINIMUM OA DAMPER AT MAXIMUM OPEN POSITION (ABOVE), BEGIN TO THROTTLE RETURN DAMPER CLOSED TO OBTAIN MINIMUM OA SETPOINT. 4. PROVIDE A MANUAL OVERRIDE OVERRIDING DAMPER CONTROL FROM GRAPHIC INTERFACE WITH INDICATOR WHEN IN MANUAL OVERRIDE MODE. ECONOMIZER TEMPERATURE CONTROL 1. THE CONTROLLER SHALL MODULATE THE ECONOMIZER, RELIEF AND RETURN DAMPERS IN SEQUENCE TO MAINTAIN THE MIXED AIR TEMPERATURE SETPOINT 2°F (ADJ.) LESS THAN DISCHARGE AIR TEMPERATURE SETPOINT. 2. THE ECONOMIZER SHALL BE ENABLED WHENEVER: a. OUTSIDE AIR TEMPERATURE IS LESS THAN 60°F (ADJ.). b. AND THE OUTSIDE AIR ENTHALPY IS LESS THAN THE RETURN AIR ENTHALPY. c. AND THE SUPPLY FAN STATUS IS ON. d. THE PREHEAT CONTROL OUTPUT HAS BEEN OFF CONTINUOUSLY FOR AT LEAST 10 MINUTES (ADJ.). e. WHEN THE UNIT IS STARTED IN OCCUPIED MODE AND OUTSIDE AIR TEMPERATURE IS LESS THAN 40°F (ADJ.), THE MINIMUM OUTSIDE AIR (OA) DAMPER SHALL RAMP OPEN TO ITS MINIMUM SETPOINT OVER A 5 MINUTE (ADJ.) PERIOD AND THE ECONOMIZER PID CALCULATION SHALL BE DISABLED ONLY AFTER THIS PERIOD SHALL THE ECONOMIZER PID OUTPUT START CALCULATING. WHEN OAT IS <40°F, THERE SHALL NOT BE A DELAYED RAMP OPEN PERIOD. THIS IS TO PREVENT THE ECONOMIZER DAMPER FROM AUTOMATICALLY DRIVING OPEN TOO QUICKLY WHILE TRYING TO SATISFY DAT SETPOINT AND TRIPPING AND RETRIPPING THE FREEZE/STAT. 3. WHEN ECONOMIZER IS ENABLED, THE FOLLOWING OPERATIONS SHALL BE ALLOWED TO OCCUR AS NEEDED AND IN THE ORDER LISTED IN ORDER ACHIEVE DISCHARGE AIR TEMPERATURE SETPOINT: a. FIRST, MINIMUM OA DAMPER SHALL MODULATE OPEN BEYOND THE POSITION REQUIRED FOR MINIMUM VENTILATION AIR AND UP TO FULLY OPEN. (FIRST 10% OF ECONOMIZER PID OUTPUT SHALL DRIVE MINIMUM OA DAMPER FULLY OPEN IF NOT ALREADY). b. THEN, ECONOMIZER DAMPER SHALL MODULATE FROM CLOSED TO FULLY OPEN. 10-55% ECONOMIZER PID OUTPUT SHALL OPEN ECONOMIZER DAMPER FROM 0-100% OPEN. c. THEN RETURN AIR DAMPER SHALL MODULATE TO FULLY CLOSED. 55-100% ECONOMIZER PID OUTPUT SHALL CONTINUE TO RESET RETURN DAMPER FROM ITS CONTROLLED MODULATED POSITION TO ACHIEVE MINIMUM OA AND PLENUM PRESSURES TO FULLY CLOSED. 0% OPEN. d. THE RELIEF AIR FANS SHALL OPERATE AS DESCRIBED IN THEIR RELATED SEQUENCE OF OPERATION. 4. THE ECONOMIZER DAMPER SHALL CLOSE AND THE RETURN AIR DAMPER SHALL OPEN WHEN: a. MIXED AIR TEMPERATURE DROPS FROM 50°F (ADJ.) TO 45°F (ADJ.). b. OR THE FREEZE/STAT IS ON. c. OR LOSS OF SUPPLY FAN ON STATUS. d. OR IF UNIT IS COMMANDED OFF. 5. DURING UNOCCUPIED MODE, THE ECONOMIZER SEQUENCE OF OPERATIONS SHALL BE DISABLED.	

TEMPERATURE AND HUMIDITY CONTROL	
GENERAL 1. GENERAL: THE CONTROLLER SHALL MONITOR THE FOLLOWING TEMPERATURE/HUMIDITY SENSORS AND USE AS REQUIRED FOR CONTROL FUNCTIONS AS DESCRIBED IN THE SECTIONS FOLLOWING. a. OUTSIDE AIR TEMPERATURE. b. OUTSIDE AIR RELATIVE HUMIDITY (USED AS REQUIRED FOR OA ENTHALPY CALCULATION) c. RETURN AIR TEMPERATURE. i. ALARMS SHALL BE PROVIDED AS FOLLOWS: 1) HIGH RETURN AIR TEMP: IF THE RETURN AIR TEMPERATURE IS GREATER THAN 80°F (ADJ.) FOR MORE THAN 30 MINUTES (ADJ.). 2) LOW RETURN AIR TEMP: IF THE RETURN AIR TEMPERATURE IS LESS THAN 60°F (ADJ.) FOR MORE THAN 30 MINUTES (ADJ.). d. RETURN AIR HUMIDITY: THE CONTROLLER SHALL MONITOR THE LOCAL AIR HUMIDITY AND USE AS REQUIRED (ENTHALPY CALCULATION) FOR ECONOMIZER CONTROL. i. ALARMS SHALL BE PROVIDED AS FOLLOWS: 1) HIGH RETURN AIR HUMIDITY: IF THE RETURN AIR HUMIDITY IS GREATER THAN UPPER LIMIT OF 70% (ADJ.). 2) LOW RETURN AIR HUMIDITY: IF THE RETURN AIR HUMIDITY IS GREATER THAN LOWER LIMIT OF 30% (ADJ.). e. MIXED AIR TEMPERATURE: THE CONTROLLER SHALL MONITOR THE MIXED AIR TEMPERATURE AND USE AS REQUIRED FOR ECONOMIZER CONTROL, OR PREHEATING CONTROL. i. THE LOW LIMIT MIXED AIR TEMPERATURE SENSOR SHALL BE AN AVERAGING TYPE OF SUFFICIENT LENGTH AND PROPERLY ARRANGED IN FURTHEST DOWNSTREAM SECTION OF MIXED AIR PLENUM AFTER AIR BLENDING APPARATUS IN ORDER TO COMPENSATE FOR INCOMPLETE MIXING AND TEMPERATURE VARIATIONS OF THE MIXED AIR STREAM. THE LENGTH OF THE SENSOR SHALL BE GREATER OF 20 MINUTES OR 1 PER 30 FT. OF PREHEAT COIL FACE AREA. ii. ALARMS SHALL BE PROVIDED AS FOLLOWS: a. HIGH MIXED AIR TEMP: ONLY WHEN ECONOMIZER IS ENABLED. IF THE MIXED AIR TEMPERATURE IS GREATER THAN 5°F (ADJ.) ABOVE EFFECTIVE MAT SETPOINT OR UPPER LIMIT OF 85°F (ADJ.). b. LOW MIXED AIR TEMP: ONLY WHEN ECONOMIZER IS ENABLED. IF THE MIXED AIR TEMPERATURE IS LESS THAN 5°F (ADJ.) BELOW EFFECTIVE MAT SETPOINT OR LOWER LIMIT OF 40°F (ADJ.). f. PREHEAT LEAVING AIR TEMP: SEE PREHEAT TEMPERATURE CONTROL. g. COOLING COIL LEAVING AIR TEMP: SEE COOLING TEMPERATURE CONTROL. h. DISCHARGE AIR TEMPERATURE: i. ALARMS SHALL BE PROVIDED AS FOLLOWS: a) HIGH DISCHARGE AIR TEMP: IF GREATER THAN 5°F (ADJ.) ABOVE EFFECTIVE DAT SETPOINT OR UPPER LIMIT OF 75°F (ADJ.). b) LOW DISCHARGE AIR TEMP: IF LESS THAN 5°F (ADJ.) BELOW EFFECTIVE DAT SETPOINT OR LOWER LIMIT OF 50 FOR RECORD OR 45°F (ADJ.) FOR 100%. ii. CHILLED WATER RETURN TEMP: SEE COOLING COIL CONTROL AND MISCELLANEOUS ALARMS AND SYSTEM DIGNOSTICS 2. AHU HW / CHW REQUESTS SHALL BE INITIALLY DEFINED AS FOLLOWS: a. ANY HW OR CHW CONTROL VALVE PID OUTPUT MUST BE GREATER THAN 99% (ADJ.) BEFORE REQUESTS ARE GENERATED, WITH NO MORE THAN 5% (ADJ.) HYSTERESIS. b. REQUEST ANALOG VALVE SETPOINT IS CALCULATED AS THE DIFFERENCE OF THE CONTROLLED TEMPERATURE FROM ITS SETPOINT, MULTIPLIED BY A SENSITIVITY FACTOR OF 2 (ADJ.). c. INCLUDE AN UPPER VALUE LIMIT OF 0 (ADJ.) REQUESTS PER INDIVIDUAL CONTROL VALVE. 3. DISCHARGE AIR TEMPERATURE SETPOINT - OPTIMIZED: THE CONTROLLER SHALL MONITOR THE DISCHARGE AIR TEMPERATURE AND SHALL RESET THE DISCHARGE AIR TEMPERATURE SETPOINT BASED ON OUTSIDE AIR CONDITIONS AND SATISFYING ALL ZONE COOLING REQUIREMENTS. a. THE INITIAL DISCHARGE AIR TEMPERATURE SETPOINT (DATSPT) SHALL BE 55°F (ADJ.). b. THE DATSPT SHALL INCREMENTALLY RESET BETWEEN A HIGH AND LOW OFFSET RANGE AS DESCRIBED BELOW: AS ZONE COOLING DEMAND INCREASES, A MINIMUM OF 1 COOLING REQUESTS (ADJ.) THE SETPOINT SHALL INCREMENTALLY RESET DOWN TO A MINIMUM OF THE LOW OFFSET. WITH NO COOLING REQUESTS, THE SETPOINT SHALL INCREMENTALLY RESET UP TO A MAXIMUM HIGH OFFSET. i. COOLING REQUESTS SHALL BE DETERMINED AT THE TERMINAL UNIT ZONE LEVEL. ii. INCREMENT SHALL BE 0.2°F/MINUTE (ADJ.). c. WHEN OUTSIDE AIR TEMPERATURE IS ABOVE 70°F (ADJ.), THE DATSPT SETPOINT SHALL RANGE FROM HIGH OFFSET OF +5°F (ADJ.) TO A LOW OFFSET OF -5°F (ADJ.). d. WHEN OUTSIDE AIR DRY BULB TEMPERATURE IS BETWEEN 70°F AND 80°F (ADJ.), THE DATSPT SHALL RANGE FROM HIGH OFFSET OF +5°F (ADJ.) TO A LOW OFFSET OF 0°F (ADJ.). e. WHEN OUTSIDE AIR WET BULB TEMPERATURE IS BELOW 60°F (ADJ.), THE DATSPT SETPOINT SHALL RANGE FROM HIGH OFFSET OF +10°F (ADJ.) TO A LOW OFFSET OF 0°F (ADJ.). DEHUMIDIFICATION CONTROL: 1. THE RETURN AIR HUMIDITY SENSOR H-3 SHALL OVERRIDE THE COOLING COIL CONTROLS AND MODULATE THE COOLING COIL VALVE TO PROVIDE DEHUMIDIFICATION. THE ZONE TEMPERATURE SENSORS (VAV ZONE REHEAT THERMOSTATS) SHALL THEN CONTROL THE VAV REHEAT VALVES TO MAINTAIN THEIR RELATIVE SPACE TEMPERATURE SETPOINTS AS FOLLOWS: 2. THE SET-POINT FOR DEHUMIDIFICATION SHALL BE 50%RH (ADJUSTABLE) AT THE RETURN AIR STREAM. WHEN THE RETURN AIR RELATIVE HUMIDITY REACHES 50%RH (ADJUSTABLE), CONTROL OF AHU THE CHILLED WATER VALVE SHALL BE RETURNED TO NORMAL OPERATION. COOLING COIL CONTROL 1. THE CONTROLLER SHALL MONITOR THE UNIT DISCHARGE AIR TEMPERATURE SENSOR AND USE AS REQUIRED FOR COOLING COIL CONTROL. 2. THE CHW VALVE SHALL MODULATE LAST IN SEQUENCE TO MAINTAIN THE UNIT DAT SETPOINT. THE COOLING COIL CONTROL SHALL BE ENABLED WHENEVER: a. DISCHARGE AIR TEMPERATURE (DAT) EXCEEDS EFFECTIVE DAT SETPOINT (OR GENERAL COOLING REQUEST). b. AND OUTSIDE AIR TEMPERATURE IS GREATER THAN EITHER THAN THE EFFECTIVE DAT SETPOINT - 2°F (ADJ.) OFFSET FOR FAN HEAT, WITH A LOWER LIMIT OF 54°F (ADJ.). c. AND THE ECONOMIZER IS EITHER DISCHARGED OR FULLY OPEN. d. AND THE SUPPLY FAN STATUS IS ON. e. AND PREHEAT DAT SETPOINT IS LESS THAN 70°F (ADJ.). f. THE CHW VALVE SHALL OPEN TO 50% ON ACTIVATION OF THE FREEZE/STAT WITH FANS OFF. 3. COOLING COIL LEAVING AIR TEMPERATURE (CCLAT): THE CONTROLLER SHALL ALSO MONITOR THE COOLING COIL LEAVING AIR TEMPERATURE AND USE AS REQUIRED FOR DIGNOSTICS AS DESCRIBED. PREHEAT TEMPERATURE CONTROL THE PREHEAT COIL SHALL BE CONTROLLED AS FOLLOWS: 1. THE CONTROLLER SHALL MODULATE THE PHC VALVE TO MAINTAIN THE DAT HEATING SETPOINT, NORMALLY LIMITED TO 50°F (ADJ.). 2. THE HEATING SETPOINT SHALL BE ABLE TO BE USER SELECTABLE TO AUTOMATICALLY RESET AS FOLLOWS: a. AS DAT DROPS FROM 40°F (ADJ.) TO 0°F (ADJ.), THE DAT HEATING SETPOINT SHALL RESET UPWARDS FROM 55°F (ADJ.) UP TO 70 MAX°F (ADJ.). 3. THE PREHEATING VALVE SHALL BE ENABLED WHENEVER: a. OUTSIDE AIR TEMPERATURE IS LESS THAN 52°F (ADJ.). b. AND THE SUPPLY FAN STATUS IS ON. c. AND THE COOLING IS NOT ACTIVE. d. AND ECONOMIZER IS NOT ACTIVE (OA DAMPER IS AT MINIMUM POSITION AND MAT IS BELOW DATSETPOINT). e. OR THE UNIT IS OFF AND OAT IS BELOW 40 DEGREES (SEE BELOW). 3. PREHEAT DISCHARGE AIR TEMPERATURE (PHDAT): THE CONTROLLER SHALL MONITOR THE PREHEAT COIL LEAVING AIR TEMPERATURE. a. WHEN UNIT IS STOPPED, AND OAT IS LESS THAN 40 DEGREES, THE PREHEAT PHDAT SHALL MAINTAIN PHDAT AT 40 DEGREES (ADJ.), WITH A LOW LIMIT ALARM AT 35 DEGREES. b. PHDAT SHALL ALSO BE USED FOR SYSTEM DIGNOSTIC / COMMISSIONING ALARMS AS DESCRIBED IN MISCELLANEOUS ALARM SEGMENT. 4. THE HOT WATER RECIRCULATION PUMP SHALL BE ON AND THE PHC VALVE SHALL FULLY OPEN TO COOL FOR FREEZE PROTECTION WHENEVER: a. DISCHARGE AIR TEMPERATURE DROPS FROM 45°F (ADJ.) TO 35°F (ADJ.). b. OR ON ACTIVATION OF THE FREEZE/STAT. 5. ALARMS SHALL BE PROVIDED AS FOLLOWS: a. LOW PREHEAT FAULT: IF THE PREHEAT IS ENABLED (ECONOMIZER AND COOLING DEMAND), AND DAT IS 5°F (ADJ.) LESS THAN DAT HEATING SETPOINT FOR MORE THAN 5 MINUTES (ADJ.) CONTINUOUS. b. HIGH TEMP ALARM: IF GREATER THAN 5°F (ADJ.) ABOVE EFFECTIVE PHDAT SETPOINT OR UPPER LIMIT OF 75°F (ADJ.), BUT ONLY WHEN MAT IS LESS THAN PHDAT SETPOINT. DEMAND CONTROL VENTILATION: 1. IN THE OCCUPIED MODE, THE BMS SHALL CONTROL THE OUTSIDE AIR FLOW AT THE AHU AS FOLLOWS: a. A MINIMUM OUTSIDE AIR FLOW OF 4,000CFM (ADJUSTABLE) FOR THE AHU SHALL BE MAINTAINED BY MODULATING THE MINIMUM OUTSIDE AIR FLOW DAMPER D-5. b. THE BMS SHALL COMPARE OUTDOOR AIR CO2 LEVELS WITH THE INDOOR AIR CO2 LEVELS MEASURED BY ANY OF THE ZONE CO2 SENSORS. c. IF A CONCENTRATION ABOVE THE OUTDOOR AIR CO2 LEVELS OF 200PPM (ADJUSTABLE) IS REACHED IN ANY OF THE ZONES, THE OUTSIDE AIR DAMPER SHALL MODULATE OPEN SUCH THAT ZONE CO2 LEVELS SHALL NOT EXCEED 300PPM (ADJUSTABLE) ABOVE OUTSIDE AIR CO2 LEVELS. d. IN THE EVENT THAT THE OUTDOOR AIR CO2 LEVELS CONTINUE TO INCREASE, THE OUTSIDE AIR DAMPER SHALL MODULATE OPEN TO MAINTAIN THE DESIGN MI OUTSIDE AIR FLOW. 2. POINT ALARMS: THE FOLLOWING POINT ALARMS SHALL BE GENERATED AT THE OPERATOR WORK STATION. a. HIGH CARBON DIOXIDE CONCENTRATION: IF THE HIGHEST ZONE CO2 CONCENTRATION IS GREATER THAN 10% (ADJ.) ABOVE SETPOINT FOR MORE THAN 15 MIN (ADJ.) WITH CRITICAL ZONE MINIMUM AIRFLOW RESET TO MAX AND OA DAMPER AT FULL DESIGN MINIMUM. ENTHALPY WHEEL CONTROL - OCCUPIED AND UNOCCUPIED MODE THE BMS SHALL CONTROL THE ENERGY RECOVERY ENTHALPY WHEEL AS FOLLOWS: 1. THE ENERGY RECOVERY ENTHALPY WHEEL SHALL RUN ONLY WHEN THE AHU IS IN THE OCCUPIED MODE. 2. THE ENERGY RECOVERY ENTHALPY WHEEL SHALL BE CONTROLLED AS DESCRIBED BELOW: a. WHEN OUTDOOR ENTHALPY IS MORE THAN 26.1 BTU/LB (ADJ.), EQUIVALENT OF 75.0°F AND 55% RH, THE BMS SHALL ROTATE THE ENTHALPY WHEEL AT FULL SPEED. b. BYPASS DAMPERS D-2 AND D-5 SHALL BE FULLY CLOSED. c. WHEN OUTDOOR ENTHALPY IS LESS THAN 26.1 BTU/LB (ADJ.), EQUIVALENT OF 75.0°F AND 55% RH, THE BMS SHALL STOP THE ENTHALPY WHEEL. i. BYPASS DAMPERS D-4 AND D-5 SHALL BE FULLY OPEN. ii. IF THE BMS SENSING THAT THE MIXED AIR TEMPERATURE, AS SENSED BY THE TEMPERATURE SENSOR LOCATED JUST UPSTREAM OF THE OF THE PREHEAT COIL, IS AT OR BELOW 55.0°F (ADJ.) THAN THE BMS SHALL ROTATE THE ENTHALPY WHEEL AT FULL SPEED (USE THE ENTHALPY WHEEL AS FIRST STAGE OF PREHEAT) TO MAINTAIN THE AHU DISCHARGE TEMPERATURE SETPOINT. i. WITH THE WHEEL AT FULL SPEED, IF THE BMS IS SENSING THAT THE AHU DISCHARGE TEMPERATURE IS INCREASING ABOVE SETPOINT, THAN IT SHALL MODULATE OPEN BYPASS DAMPERS D-2 AND D-5 TO MAINTAIN THE AHU DISCHARGE TEMPERATURE SETPOINT. ii. THE PREHEAT COIL CONTROL VALVE SHALL BE CLOSED. iii. WITH THE WHEEL AT FULL SPEED AND WHEEL BYPASS DAMPERS FULLY CLOSED, IF THE BMS IS SENSING THAT THE AHU DISCHARGE TEMPERATURE IS INCREASING ABOVE SETPOINT, THAN THE BMS SHALL STOP THE ENTHALPY WHEEL. iv. THE PREHEAT COIL CONTROL VALVE SHALL MODULATE AS REQUIRED TO MAINTAIN THE AHU DISCHARGE TEMPERATURE SETPOINT. iii. WITH THE WHEEL AT FULL SPEED AND WHEEL BYPASS DAMPERS FULLY CLOSED, IF THE BMS IS SENSING THAT THE AHU DISCHARGE TEMPERATURE IS DECREASING ABOVE SETPOINT, THAN IT SHALL MODULATE OPEN THE PREHEAT COIL CONTROL VALVE AS REQUIRED TO MAINTAIN THE AHU DISCHARGE TEMPERATURE SETPOINT. 3. WITH THE WHEEL AT FULL SPEED AND IF THE RELIEF AIR TEMPERATURE REACHES THE FROST CONTROL SETPOINT (DEFAULT 34.0°F), THEN THE BMS SHALL STOP THE WHEEL IN ORDER TO AVOID ICE FORMATION. a. WITHIN THE WHEEL'S MEDIA. b. THE BMS SHALL WAIT FOR A PERIOD OF NO LESS THAN 30 MIN. (ADJ.) BEFORE RESTARTING THE WHEEL. FREEZE PROTECTION 1. IF THE MIXED AIR TEMPERATURE SENSOR FALLS THE OUTSIDE AIR DAMPER SHALL CLOSE AND AN ALARM SHALL BE ANNUNCIATED AT THE BAS. 2. A HARDWIRED, LOW LIMIT TEMPERATURE SWITCH SHALL BE ELECTRICALLY INTERLOCKED WITH THE VARIABLE SPEED DRIVE OF EACH FAN. 3. IF THE LOW LIMIT TEMPERATURE SWITCH IS TRIPPED 30.0 DEG. F (ADJ.), THE OUTSIDE AIR DAMPER SHALL CLOSE AND AN ALARM SHALL BE ANNUNCIATED AT THE BMS. A MANUAL RESET OF THE LOW LIMIT TEMPERATURE SWITCH SHALL BE REQUIRED TO RESTART THE FANS. FILTER STATUS 1. A DIFFERENTIAL PRESSURE SWITCH SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE FILTERS WHEN THE FAN IS RUNNING. IF THE SWITCH CLOSURES DURING NORMAL OPERATION A DIRTY FILTER ALARM SHALL BE ANNUNCIATED. FIRE ALARM SHUTDOWN CONTROL 1. THE SUPPLY AND RELIEF FANS SHALL BE INTERLOCKED WITH THE FIRE ALARM SYSTEM THROUGH A CONTROL MODULE (CM). THE CONTROL MODULE SHALL BE HARD WIRED INTERLOCKED TO CONTROL THE FANS IN THE HAND AND AUTO OPERATING MODES. 2. THE SMOKE DETECTORS LOCATED INSIDE OF THE UNIT SHALL BE MONITORED BY THE FIRE ALARM SYSTEM. THE FIRE ALARM SYSTEM SHALL BE PROGRAMMED TO ACTIVATE THE CONTROL MODULE AND THE SYSTEM SHALL OPERATE AS FOLLOWS: a. WHEN PRODUCTS OF COMBUSTION ARE SENSED BY THE SMOKE DETECTORS, THE CONTROL MODULE SHALL STOP THE FANS IN THE HAND AND AUTO OPERATING MODES. b. WHEN THE PRODUCTS OF COMBUSTION ARE CLEARED AND THE FIRE ALARM SYSTEM IS RESET, THE CONTROL MODULE SHALL RESTORE NORMAL CONTROL TO THE FANS. FAILURE MODE: 1. UPON LOSS OF CONTROL SIGNAL OR ELECTRICAL POWER THE CONTROL DEVICES SHALL FAIL IN THE MANNER INDICATED IN THE "BMS POINT FUNCTION SCHEDULE" ON THE DRAWINGS. SET POINT ADJUSTMENTS: 1. ALL SETPOINTS SHALL BE USER ADJUSTABLE THROUGH THE BMS AT THE OWS. LEED IAQ PRE-OCCUPANCY FLUSH-OUT - SEQUENCE OF OPERATION 1. PRIOR TO OCCUPANCY, IN ORDER TO COMPLY WITH LEED EQ CREDIT 3.2, THE AHU SHALL RUN @ 100% OUTSIDE AIR FOR MINIMUM 10 DAYS NONSTOP (24/7). ALL VAV'S SHALL BE COMMANDED FULLY OPENED. ALL EXHAUST FANS SHALL BE COMMANDED OFF. b. BYPASS THE HEAT WHEEL, MODULATE CHILLED WATER AND/OR HEATING HOT WATER CONTROL VALVES TO MAINTAIN TEMPERATURE AT T-8 TO BE GREATER THAN 80°F (ADJ.) AND LESS THAN 80°F (ADJ.). c. CHILLED WATER AND/OR HEATING HOT WATER CONTROL VALVES AT THE AHU SHALL BE OVERRIDDEN IF HUMIDITY PROBE H-3 DETECTS HUMIDITY GREATER THAN 55% (ADJ.) OR LESS THAN 40% RH (ADJ.). d. CHILLED WATER AND/OR HEATING HOT WATER CONTROL VALVES SHALL MODULATE TO ENSURE HUMIDITY NEVER CLIMBS ABOVE 80% RH. e. THERE SHALL BE NO MIXING - ALL RELIEF AIR SHALL BE DISCHARGED OUTSIDE OF THE BUILDING. THIS SEQUENCE SHALL BE INVISIBLE TO MAINTENANCE PERSONNEL AND ONLY ACCESSIBLE TO THE CONTROLS CONTRACTOR.	

MISCELLANEOUS ALARMS & SYSTEMS DIAGNOSTICS	
 1. WHEN THE ASSOCIATED AHU FAN STATUS IS ON, THE FOLLOWING MONITORING AND ALARM FUNCTIONS SHALL INITIATE ONE OF SEVERAL PRIMARY CATEGORIES OF ALARMS DISPLAYED AT THE BMS WORKSTATION. AT THE EQUIPMENT CONTROLLER LEVEL, EACH ALARM WITHIN EACH CATEGORY SHALL BE LABELED INDEPENDENTLY FOR EASY DIAGNOSTIC PURPOSES. PRIMARY CATEGORIES SHALL BE AS FOLLOWS IN BOLD: a. FAN CONTROL FAULT i. UNSTABLE PID LOOP: IF ANY FAN SPEED PID LOOP CONTINUES TO CYCLE ITS OUTPUT MORE THAN 40% OF ITS RANGE (ADJ.) 3 TIMES (ADJ.) IN ANY 60 MINUTE INTERNAL. ii. "EXCESSIVE SUPPLY FAN SPEED" ALARM: IF THE SUPPLY FAN SPEED OUTPUT REMAINS ABOVE 95% FOR MORE THAN 8 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 3 (ADJ.) CONSECUTIVE OCCUPIED PERIODS. iii. "EXCESSIVE RELIEF FAN SPEED" ALARM: IF THE RELIEF FAN SPEED OUTPUT REMAINS ABOVE 95% FOR MORE THAN 8 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 3 (ADJ.) CONSECUTIVE OCCUPIED PERIODS. iv. CONTINUOUS ZONE "CHECK SIZING" AIRFLOW REQUEST ALARM: IF ONE OR MORE ZONE TERMINAL UNIT CONTROLLERS IS IN "CHECK SIZING" ALARM MODE FOR COOLING (SENDING CONTINUOUS AIRFLOW REQUESTS), AND DUCT STATIC PRESSURE HAS RESET TO MAXIMUM SETPOINT, AND THE COMBINED CONDITION REMAINS FOR MORE THAN 8 HOURS (ADJ.) CONTINUOUSLY. v. AIRFLOW SIGNAL OUT OF RANGE: IF MEASURED SUPPLY AND/OR RELIEF AIRFLOW VARIES SIGNIFICANTLY FROM THE INITIAL BASELINE FANSETPOINT NON-LEAK OPERATING CURVE. 1) THE INITIAL BASELINE OPERATING CURVE SHALL BE DETERMINED EMPIRICALLY USING PROPERLY CALIBRATED AND MEASURED FAN SPEED, CFM READINGS AND STATIC PRESSURES DURING SYSTEM BALANCING/COMMISSIONING AND MODELED IN THE CONTROLLER BETWEEN ACTUAL OPERATING LIMITS OF MINIMUM AND MAXIMUM VFD SPEEDS FOR EACH SYSTEM IN A MAXIMUM OF 10% SPEED INCREMENTS. 2) WHEN FAN SYSTEMS ARE ENABLED, COMPARE THE ASSOCIATED AIRFLOW OUTPUT, THE DUCT SP OUTPUT, AND THE VFD SPEED OUTPUT TO THE INITIAL ACTUAL PERFORMANCE CURVE. A COMMISSIONING LEVEL ALARM SHALL BE SENT WHEN CURRENT OPERATING CONDITIONS VARY FROM BASELINE CURVE BY MORE THAN 20% (ADJ.) FOR MORE THAN 4 HOURS (ADJ.) CONTINUOUSLY. vi. VFD LEFT IN LOCAL MANUAL OPERATION: NETWORK WITH VFD INTERFACE AND ALARM IF VFD IS LOCALLY LEFT IN MANUAL SPEED CONTROL MODE OR IF VFD SPEED OUTPUT SIGNAL REMAINS CONSTANT FOR MORE THAN 12 HOURS (ADJ.) CONTINUOUSLY. c. DAMPER CONTROL FAULT i. UNSTABLE PID LOOP: IF ANY DAMPER PID LOOP CONTINUES TO CYCLE ITS OUTPUT MORE THAN 40% OF ITS RANGE (ADJ.) 3 TIMES (ADJ.) IN ANY 60 MINUTE INTERNAL. ii. OUTDOOR AIR DAMPER CLOSE OFF FAILURE: IF MIXED AIR TEMPERATURE IS EITHER LESS THAN OR GREATER THAN RETURN AIR TEMPERATURE BY MORE THAN 3°F (ADJ.) WHEN MINIMUM OA DAMPER OUTPUT IS AT 0% OUTPUT FOR A MINIMUM 30 MINUTES (ADJ.) CONTINUOUSLY. iii. EXCESSIVE DEMAND BASED VENTILATION DEMAND: IF DUE TO DEMAND BASED VENTILATION CO2 CONTROL, THE MINIMUM OA RESETS TO DESIGN (MAXIMUM MINIMUM OA AIRFLOW SETPOINT AND REMAINS IN THAT CONDITION FOR MORE THAN 8 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 3 (ADJ.) CONSECUTIVE OCCUPIED PERIODS. d. HEATING/COOLING FAULT i. UNSTABLE PID LOOP: IF ANY HEATING OR COOLING PID LOOP CONTINUES TO CYCLE ITS OUTPUT MORE THAN 40% OF ITS RANGE (ADJ.) 3 TIMES (ADJ.) IN ANY 60 MINUTE INTERNAL. ii. IF THE DAT SETPOINT RESETS AND REMAINS AT MINIMUM (DEHUMIDIFICATION MODE) FOR MORE THAN 6 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 3 (ADJ.) CONSECUTIVE OCCUPIED PERIODS. 2) OR IF COOLING VALVE OUTPUT REMAINS ABOVE 95% FOR MORE THAN ACCUMULATED 8 HOURS (ADJ.) PER OCCUPIED PERIOD FOR AT LEAST 3 (ADJ.) CONSECUTIVE OCCUPIED PERIODS. iii. RETURN RH SENSOR CHECK: 1) IF THE RA RH SENSOR CONTINUOUSLY READS AT LEAST 10% (ADJ.) ABOVE MAXIMUM SETPOINT (THEREBY ENABLING DEHUMIDIFICATION MODE) AND/OR IF OAT IS LESS THAN MINIMUM LOW DATSPT FOR A MINIMUM OF 8 CONTINUOUS HOURS (ADJ.). iv. TEMPERATURE SENSOR CALIBRATION DIAGNOSTIC RUN: 1) RUN UNIT AT MINIMUM FAN SPEEDS FOR A MINIMUM OF 20 MINUTES EACH SATURDAY AT 3 A.M IN 100% RECIRCULATION MODE. (RA DAMPER IS OPEN, OA AND RELIEF DAMPERS FULLY CLOSED AND WITH ALL HEATING AND COOLING CONTROLS DISABLED AND COMPARE AND TREND ALL TEMPERATURE SENSORS TO EACH OTHER. 2) ALARM IF RAT, MAT, PHDAT, CDAT, RHDAT, DAT VARY FROM EACH OTHER BY MORE THAN +/- 1°F AFTER 15 MINUTES ELAPSED TIME (ADJ.). 3) INCLUDE COMPENSATION FACTORS FOR FAN HEAT DEPENDING ON RELATIVE LOCATIONS OF SENSORS AND FAN ARRANGEMENT. COORDINATE WITH TAB AGENT TO OBTAIN THOSE FACTORS WITH FANS RUNNING AT MINIMUM SPEED. 2. VFD COMMUNICATIONS BAS INTERFACE a. THE VFD INTERFACE SHALL BE CONNECTED DIRECTLY TO THE MAIN BMS NETWORK TRUNK TO MONITOR, DISPLAY, TREND AND REPORT THE FOLLOWING MINIMUM POINTS. v. VFD INTERFACE SHALL NOT BE NETWORKED INDRECTLY TO THE MAIN BMS THROUGH AHU CONTROLLER: i. SPEED OUTPUT ii. HANDAUTO SELECTION INDICATION iii. DRIVE AMPS iv. KW (COMPARE INSTANTANEOUS VALUE, THE CONNECTED MOTOR NAMEPLATE HP/KW (CONSTANT) AND THE RATIO) v. KW/HR (INCLUDE CALCULATED ENERGY SAVINGS FROM BASELINE IF MOTOR KW AT FULL SPEED KW/RAN CONTINUOUSLY AT FULL SPEED) vi. OPERATING HOURS vii. WARNINGS viii. FAULTS	

SHEET NOTES

1. THE SEQUENCES AND DIAGRAMS OF CONTROL, SHOWN ON THIS SHEET CONTAIN THE GENERAL CONTROL REQUIREMENTS. THESE STRATEGIES WILL BE CLARIFIED AND MODIFIED THROUGH PROGRAMMING MEETINGS BETWEEN THE CONTRACTOR, OWNER AND CONSULTANT PRE-NOI PRESENTATION. AT THAT TIME INITIAL SETPOINTS AND RESET SCHEDULES WILL BE FINALIZED BEFORE PROGRAMMING. AFTER THE SYSTEM IS OPERATIONAL, TRENDS WILL BE REQUIRED TO VERIFY THE ACCURACY AND ADEQUACY OF THE SEQUENCE OF CONTROL. ADDITIONAL FINE TUNING OR CHANGES IN STRATEGY IN ORDER TO ACHIEVE DESIGN CONDITIONS SHALL BE REQUIRED BY THE CONTROLS CONTRACTOR AS DIRECTED BY THE OWNER OR ENGINEER. PROVIDE ADDITIONAL ALARMS AS REQUESTED BY THE OWNER OR ENGINEER. ALL SETPOINTS SHALL BE OPERATOR ADJUSTABLE THRU THE BMS AT THE OPERATORS WORKING STATION (OWS).



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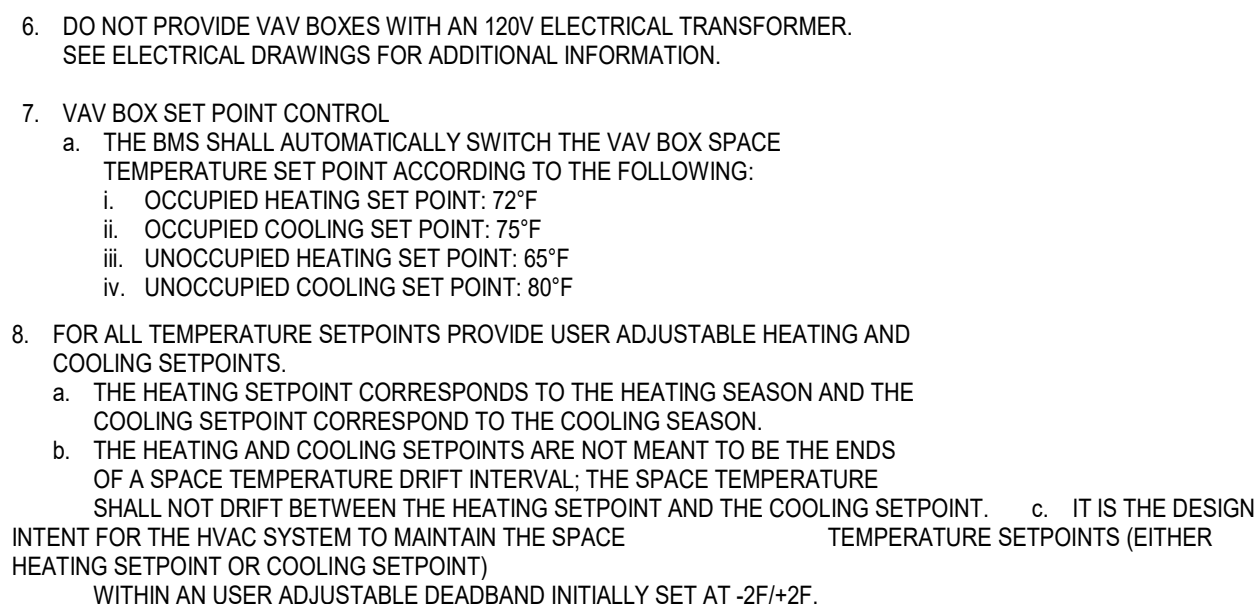
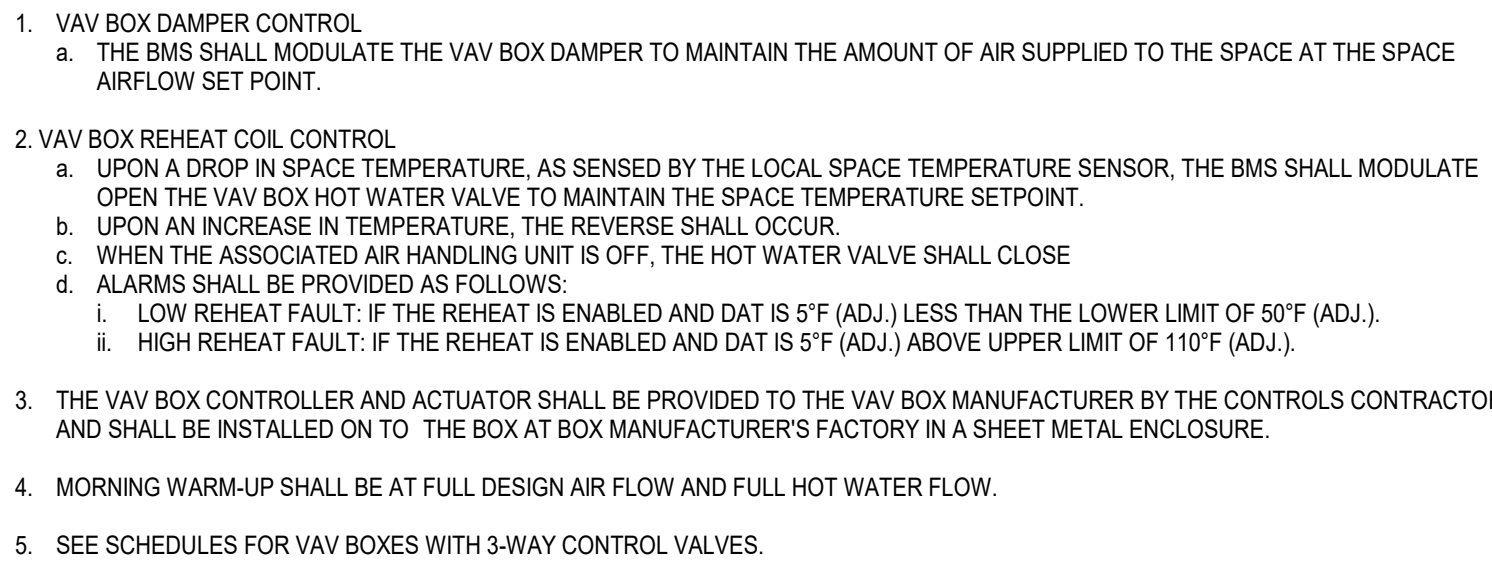
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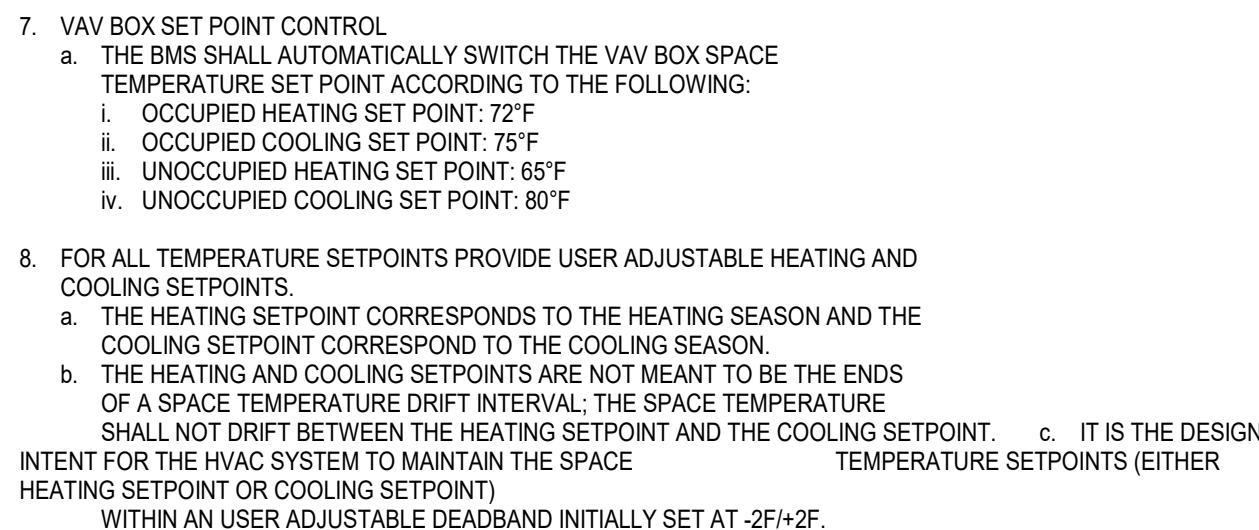
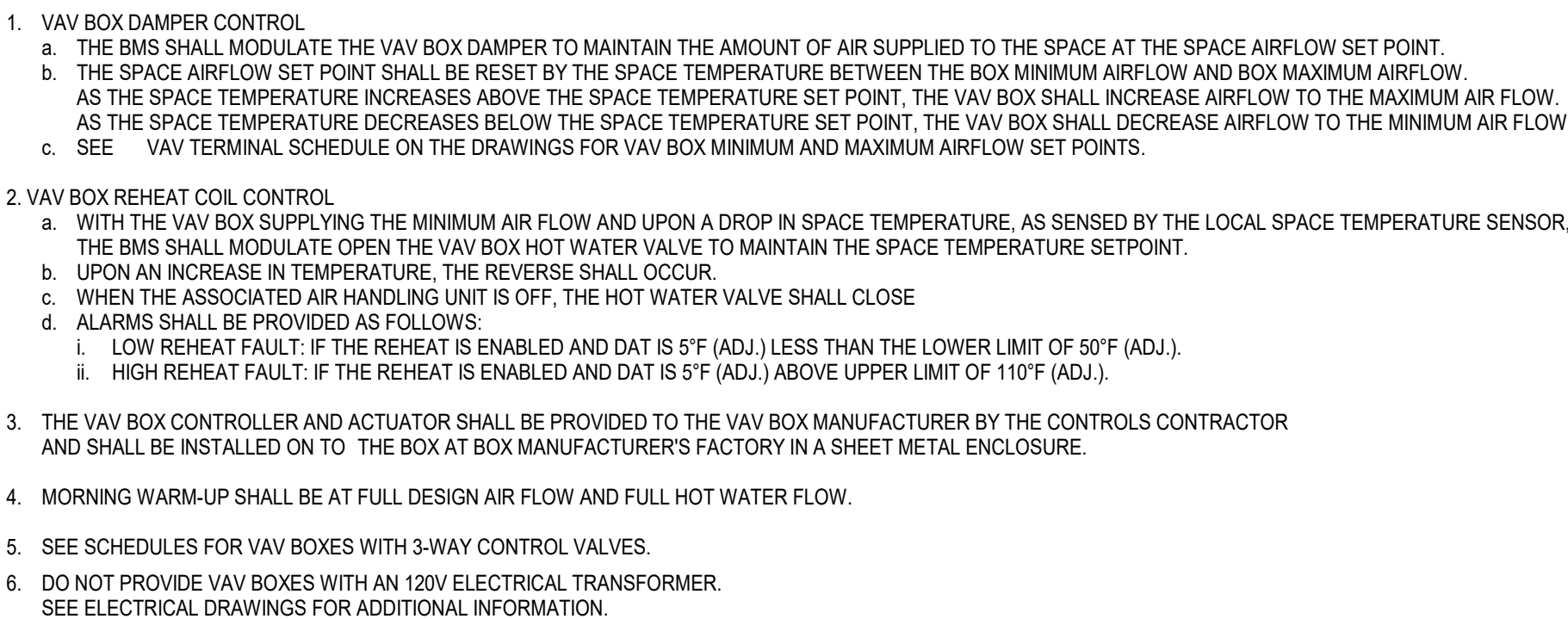
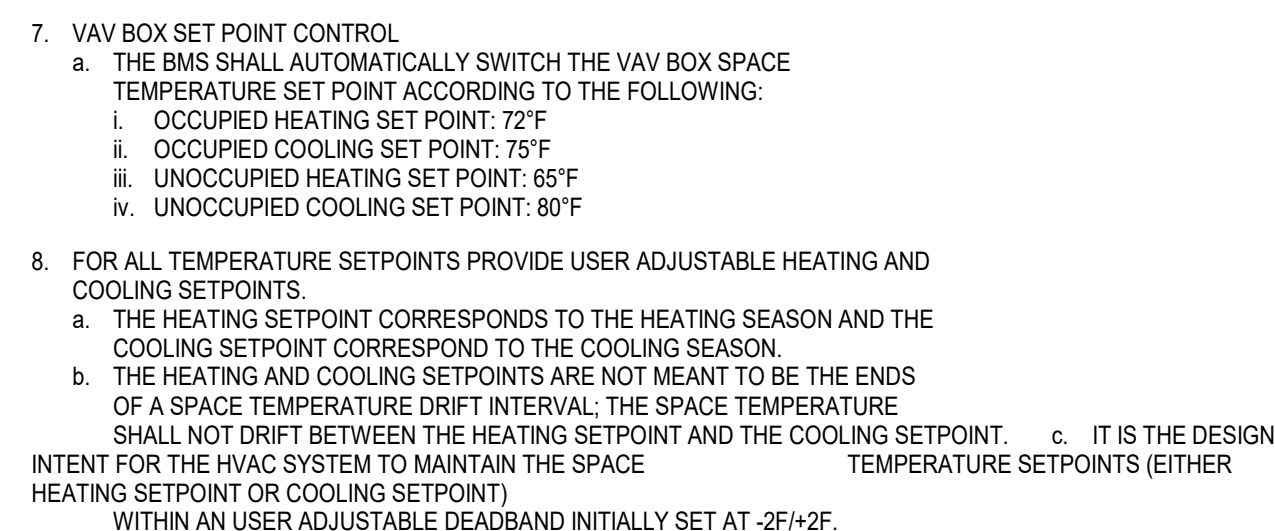
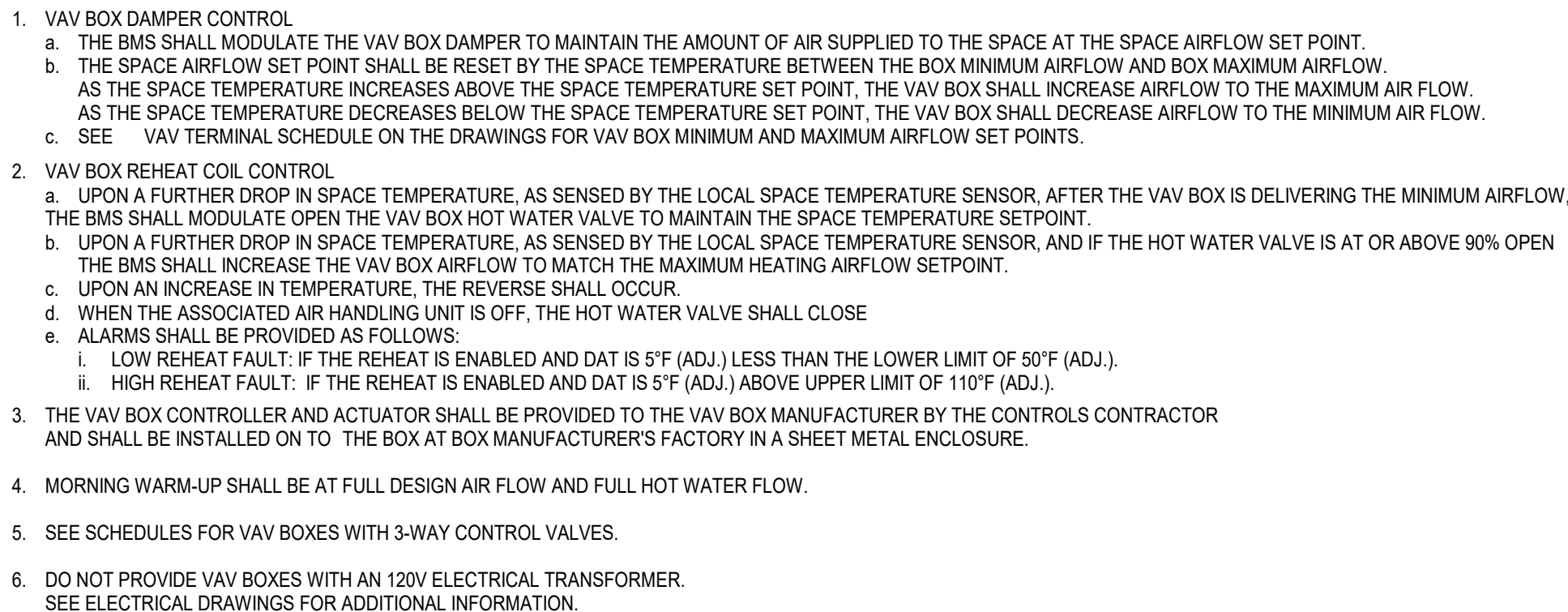
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1) CONSTANT
SCALE: NONE

2) VARIABLY
SCALE: NONE

3) SCALE: NON

CONSTANT AIR VOLUME BOX

DUAL MAXIMUM VARIABLE AIR VOLUME BOX

VARIABLE AIR VOLUME BOX

[illegible]

M8.8